

```
In [ ]: 1 import array
        2 import collections
        3 import copy
        4 import decimal
        5 import functools
        6 import importlib
        7 import logging
        8 import math
        9 import pprint
       10 import random
       11 import re
       12 import statistics
       13 import string
       14 import sys
       15 import time
       16 import timeit
       17
       18 import pandas
       19
       20
       21 num1 = 10_00_000
       22 print(num1)
```

```
1000000
```

```
In [2]: 1 print(format(0.1, ".25f"))
```

```
0.10000000000000000055511151
```

```
In [3]: 1 print(format(0.125, ".25f"))
```

```
0.125000000000000000000000000000
```

```
In [4]: 1 print(0.1 + 0.1 + 0.1 == 0.3)
```

```
False
```

```
In [5]: 1 print(format(0.3, ".25f"))
```

```
0.299999999999999999888977698
```

```
In [6]: 1 print(format(0.2, ".25f"))
```

```
0.200000000000000000111022302
```

```
In [7]: 1 print(abs((0.1 + 0.1 + 0.1) - 0.3) < 0.001)
```

```
True
```

```
In [8]: 1 print((0.125).as_integer_ratio())
```

```
(1, 8)
```

```
In [9]: 1 print((10).          (25))
```

35

```
In [10]: 1 a = float(10)
         2 print(a)
```

10.0

```
In [11]: 1 c = (-4) ** 0.5
         2 print(c)
```

(1.2246467991473532e-16+2j)

```
In [12]: 1 print(type(c))
```

<class 'complex'>

```
In [13]: 1 print(c.real)
```

1.2246467991473532e-16

```
In [14]: 1 print(c.imag)
```

2.0

```
In [15]: 1 c = 10 + 2j
         2 print(type(c))
```

<class 'complex'>

```
In [16]: 1 print(c)
```

(10+2j)

```
In [17]: 1 try:
         2     print("2" < 2)
         3 except TypeError as ex:
         4     print(type(ex).    )
```

TypeError

```
In [18]: 1 print("2" == 2)
```

False

```
In [19]: 1 a = 0
         2 b = 1
         3 c = 500
         4 d = 0
         5 e = 1
         6 f = 500
         7 print(id(a), id(b), id(c), id(d), id(e), id(f))
```

11370000 11370032 128326959784528 11370000 11370032 128326959784048

```
In [20]: 1 print(0.1 * 3 - 0.3 < abs(1e-6))
```

True

```
In [21]: 1 a = 257
        2 print(a is 257)
```

False

```
<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
<>:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
/tmp/ipykernel_188635/1958631963.py:2: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?
print(a is 257)
```

```
In [22]: 1 a = 10
        2 b = 0
        3 print(b != 0 and a / b)
```

False

```
In [23]: 1 a = 1e-6
        2 b = 2e-6
        3 tol = 1e-7
        4 print(abs(a - b) <= tol)
```

False

```
In [24]: 1 elapsed = 7835
        2 hours = elapsed // 3600
        3 minutes = (elapsed - (hours * 3600)) // 60
        4 seconds = elapsed - ((hours * 3600) + (minutes * 60))
        5 print(hours, minutes, seconds)
```

2 10 35

```
In [25]: 1 elapsed = 7835
        2 hours = elapsed // (60 * 60)
        3 minutes = (elapsed % (60 * 60)) // 60
        4 seconds = (elapsed % (60 * 60)) % 60
        5 print(hours, minutes, seconds)
```

2 10 35

```
In [26]: 1 a = None
        2 if a == None:
        3     a = "N/A"
        4 print(a)
```

N/A

```
In [27]: 1 a = None
        2 a = "N/A" if a is None else a
        3 print(a)
```

N/A

```
In [28]: 1 score = 700
        2 rating = "Invalid"
        3 if score < 0:
```

```
4         rating = "Negative"
5     elif score < 580:
6         rating = "Poor"
7     elif score < 670:
8         rating = "Fair"
9     elif score < 740:
10        rating = "Good"
11    elif score < 800:
12        rating = "Very Good"
13    elif score < 850:
14        rating = "Excellent"
15    print(rating)
```

Good

```
In [29]: 1 elapsed = 1000000
2         magnitude = "Invalid"
3         if elapsed < 60:
4             magnitude = "seconds"
5         elif elapsed < (60 * 60):
6             magnitude = "minutes"
7         elif elapsed < (60 * 60 * 24):
8             magnitude = "hours"
9         elif elapsed < (60 * 60 * 24 * 7):
10            magnitude = "days"
11        elif elapsed > (60 * 60 * 24 * 7):
12            magnitude = "weeks"
13        print(magnitude)
```

weeks

```
In [30]: 1 t = [1, 2], 3, 4
2         l = list(t)
3         print(l)
```

[[1, 2], 3, 4]

```
In [31]: 1 l[0][1] = 20
2         print(l)
```

[[1, 20], 3, 4]

```
In [32]: 1 print(t)
```

([1, 20], 3, 4)

```
In [33]: 1 for i in t:
2         print(id(i))
```

128327770840832
11370096
11370128

```
In [34]: 1 for i in l:
2         print(id(i))
```

```
128327770840832
11370096
11370128
```

```
In [35]: 1 for i, j in zip(l, t):
          2     print(i is j)
```

```
True
True
True
```

```
In [36]: 1 s = "ABCDEFGHIIJ"
          2 print(s[3:6:-1])
```

```
In [37]: 1 print(s[3:5 + 1])
```

```
DEF
```

```
In [38]: 1 m1 = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m2 = m1[0:2]
          3 print(m1[0] is m2[1])
```

```
False
```

```
In [39]: 1 m1[0] = "Python"
          2 print(m1)
```

```
['Python', [0, 0, 0], [0, 0, 0]]
```

```
In [40]: 1 print(m2)
```

```
[[0, 0, 0], [0, 0, 0]]
```

```
In [41]: 1 m1[1][0] = 100
```

```
In [42]: 1 print(m1)
```

```
['Python', [100, 0, 0], [0, 0, 0]]
```

```
In [43]: 1 print(m2)
```

```
[[0, 0, 0], [100, 0, 0]]
```

```
In [44]: 1 print("abcdef"[-1:-4])
```

```
In [45]: 1 print("abcdef"[1:4:-1])
```

```
In [46]: 1 print("racecar" == "racecar"[::-1])
```

```
True
```

```
In [47]: 1 list1 = [0, 1, 2, 3, 4]
        2 list1[0:3] = "abc"
        3 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [48]: 1 list1[0:3] = ["a", "b", "c"]
        2 print(list1)
```

```
['a', 'b', 'c', 3, 4]
```

```
In [49]: 1 list1[0:3] = ["a", "b"]
        2 print(list1)
```

```
['a', 'b', 3, 4]
```

```
In [50]: 1 del list1[2]
        2 print(list1)
```

```
['a', 'b', 4]
```

```
In [51]: 1 del list1[0:2]
        2 print(list1)
```

```
[4]
```

```
In [52]: 1 list1.extend("ABC")
        2 print(list1)
```

```
[4, 'A', 'B', 'C']
```

```
In [53]: 1 list1.insert(0, "Y0!")
        2 print(list1)
```

```
['Y0!', 4, 'A', 'B', 'C']
```

```
In [54]: 1 list1[:-3:-1] = "cb"
        2 print(list1)
```

```
['Y0!', 4, 'A', 'b', 'c']
```

```
In [55]: 1 l = []
        2 print(timeit.timeit("l.append(1)", globals = globals(), number = 100_000))
```

```
0.0030679129995405674
```

```
In [56]: 1 print(len(l))
```

```
100000
```

```
In [57]: 1 list2 = []
        2 print(timeit.timeit("list2.insert(0, 1)", globals = globals(), number = 100_000))
```

```
2.1085926070227288
```

```
In [58]: 1 print(len(list2))
```

```
100000
```

```
In [59]: 1 print(1.249788 / 0.0018043)
```

```
692.6719503408524
```

```
In [60]: 1 list2 = []  
2 print(timeit.timeit("list2.insert(len(list2), 1)", globals = globals(), number  
3 = 100_000))
```

```
0.0051941630081273615
```

```
In [61]: 1 print(0.0057634 / 0.0018043)
```

```
3.194258161059691
```

```
In [62]: 1 l1 = [[0, 1], 2, "3"]  
2 l2 = l1[:]  
3 for i in l1:  
4     print(id(i))
```

```
128326959004160  
11370064  
11434512
```

```
In [63]: 1 for i in l2:  
2     print(id(i))
```

```
128326959004160  
11370064  
11434512
```

```
In [64]: 1 l2[0][0] = "0"  
2 for i in l1:  
3     print(id(i))
```

```
128326959004160  
11370064  
11434512
```

```
In [65]: 1 for i in l2:  
2     print(id(i))
```

```
128326959004160  
11370064  
11434512
```

```
In [66]: 1 for i in l1:  
2     print(i)
```

```
['0', 1]  
2  
3
```

```
In [67]: 1 for i in l2:
         2     print(i)
```

```
['0', 1]
2
3
```

```
In [68]: 1 l2[1] = "22"
         2 for i in l1:
         3     print(id(i))
```

```
128326959004160
11370064
11434512
```

```
In [69]: 1 for i in l2:
         2     print(id(i))
```

```
128326959004160
128328432492752
11434512
```

```
In [70]: 1 for i in l1:
         2     print(i)
```

```
['0', 1]
2
3
```

```
In [71]: 1 for i in l2:
         2     print(i)
```

```
['0', 1]
22
3
```

```
In [72]: 1 l1 = [[0, 1], 2, "3"]
         2 l2 = l1[:]
         3 l2[0] = ["0", 1]
         4 for i in l1:
         5     print(id(i))
```

```
128326959032832
11370064
11434512
```

```
In [73]: 1 for i in l2:
         2     print(id(i))
```

```
128326959032704
11370064
11434512
```

```
In [74]: 1 for i in l1:
         2     print(i)
```

```
[0, 1]
2
3
```

```
In [75]: 1 for i in l2:
          2     print(i)
```

```
['0', 1]
2
3
```

```
In [76]: 1 l1 = [[0, 1], 2, "3"]
          2 l2 = l1[:]
          3 print(id(l1), id(l2))
```

```
128326959032896 128326959032640
```

```
In [77]: 1 for i, j in zip(l1, l2):
          2     print(id(i), id(j))
```

```
128326959032576 128326959032576
11370064 11370064
11434512 11434512
```

```
In [78]: 1 for i, j in zip(l1[0], l2[0]):
          2     print(id(i), id(j))
```

```
11370000 11370000
11370032 11370032
```

```
In [79]: 1 l1 = [[0, 1], 2, "3"]
          2 l2 = l1.copy()
          3 print(id(l1), id(l2))
```

```
128326959030912 128326959031744
```

```
In [80]: 1 for i, j in zip(l1, l2):
          2     print(id(i), id(j))
```

```
128326959036352 128326959036352
11370064 11370064
11434512 11434512
```

```
In [81]: 1 for i, j in zip(l1[0], l2[0]):
          2     print(id(i), id(j))
```

```
11370000 11370000
11370032 11370032
```

```
In [82]: 1 l1 = [[0, 1], 2, "3"]
          2 l2 = copy.deepcopy(l1)
          3 print(id(l1), id(l2))
```

```
128326959038016 128326959037952
```

```
In [83]: 1 for i, j in zip(l1, l2):
          2     print(id(i), id(j))
```

```
128326959038144 128326959035904
11370064 11370064
11434512 11434512
```

```
In [84]: 1 for i, j in zip(l1[0], l2[0]):
          2     print(id(i), id(j))
```

```
11370000 11370000
11370032 11370032
```

```
In [85]: 1 t1 = (0, "1", "23")
          2 t2 = t1[:]
          3 print(t1 is t2)
```

```
True
```

```
In [86]: 1 a, b, c, d = "abcd"
          2 print(a, b, c, d)
```

```
a b c d
```

```
In [87]: 1 s = "FfEeDdCcBbAa"
          2 lowerLetters = s[::-2]
          3 upperLetters = s[-2::-2]
          4 print(lowerLetters)
```

```
abcdef
```

```
In [88]: 1 print(upperLetters)
```

```
ABCDEF
```

```
In [89]: 1 t1 = 1, 2, 3, 4, 5, 6
          2 t2 = 7, 8, 9, 10
          3 t3 = 11, 12, 13, 14, 15, 16, 17
          4 t = t1 + t2 + t3
          5 l = list(t)
          6 l[::2] = [0] * len(l[::2])
          7 print(tuple(l))
```

```
(0, 2, 0, 4, 0, 6, 0, 8, 0, 10, 0, 12, 0, 14, 0, 16, 0)
```

```
In [90]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m[0][0] = 1
          3 m[1][1] = 1
          4 m[2][2] = 1
          5 print(m)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [91]: 1 m = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
          2 m1 = copy.deepcopy(m)
          3 m1[0][0] = 1
          4 m1[1][1] = 1
```

```
5 m1[2][2] = 1
6 print(m1)
```

```
[[1, 0, 0], [0, 1, 0], [0, 0, 1]]
```

```
In [92]: print(m)
1
2 data = [(100, "USD", "EUR", 0.83), (100, "USD", "CAD", 1.27), (100, "CAD",
3 "EUR", 0.65)]
4 USDAmountInEUR = data[0][0] * data[0][3]
5 print(str(data[0][0]) + " " + data[0][1] + " = " + str(USDAmountInEUR) + " " +
str(data[0][2]))
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
100 USD = 83.0 EUR
```

```
In [93]: 1 USDAmountInCAD = data[1][0] * data[1][3]
2 print(str(data[1][0]) + " " + data[1][1] + " = " + str(USDAmountInCAD) + " " +
str(data[1][2]))
```

```
100 USD = 127.0 CAD
```

```
In [94]: 1 CADAmountInEUR = data[2][0] * data[2][3]
2 print(str(data[2][0]) + " " + data[2][1] + " = " + str(CADAmountInEUR) + " " +
str(data[2][2]))
```

```
100 CAD = 65.0 EUR
```

```
In [95]: 1 print(ord("A"))
```

```
65
```

```
In [96]: 1 print("\N{SNAKE}")
```

```
🐍
```

```
In [97]: 1 print(ord("\N{SNAKE}"))
```

```
128013
```

```
In [98]: 1 print(hex(ord("\N{SNAKE}")))
```

```
0x1f40d
```

```
In [99]: 1 print("\u03b1")
```

```
α
```

```
In [100]: 1 print("\U0001f40d")
```

```
🐍
```

```
In [101]: 1 print(int("1f40d", 16))
```

```
128013
```

```
In [102]: 1 print("\N{SNAKE} is a snake")
```

🐍 is a snake

```
In [103]: 1 print(int("7ff", 16))
```

2047

```
In [104]: 1 print(int("ffff", 16))
```

65535

```
In [105]: 1 print(int("10ffff", 16))
```

1114111

```
In [106]: 1 print(bin(2047))
```

0b1111111111

```
In [107]: 1 print(len(bin(2047)))
```

13

```
In [108]: 1 print(bin(65535))
```

0b1111111111111111

```
In [109]: 1 print(len(bin(65535)))
```

18

```
In [110]: 1 print(bin(1114111))
```

0b10000111111111111111

```
In [111]: 1 print(len(bin(1114111)))
```

23

```
In [112]: 1 print(bin(128013))
```

0b11111010000001101

```
In [113]: 1 print("\n{snake}")
```

{snake}

```
In [114]: 1 a = True
          2 b = True
          3 print(a is b)
```

True

```
In [115]: 1 a = False
          2 b = False
          3 print(a is b)
```

True

```
In [116]: 1 a = True
          2 print(id(a))
```

```
11120192
```

```
In [117]: 1 a = False
          2 print(id(a))
```

```
11045056
```

```
In [118]: 1 b = False
          2 print(id(b))
```

```
11045056
```

```
In [119]: 1 ?
```

Docstring:

Return the lowest index in S where substring sub is found, such that sub is contained within S[start:end].

Optional arguments start and end are interpreted as in slice notation.
Return -1 on failure.

Type: method_descriptor

```
In [120]: 1 print(f"1 + 1 = {1 + 1}")
```

```
1 + 1 = 2
```

```
In [121]: 1 bid = 1.5760
          2 ask = 1.5763
          3 print("bid: {}, ask: {}, spread: {}".format(bid, ask, ask - bid))
```

```
bid: 1.576, ask: 1.5763, spread: 0.000299999999999996696
```

```
In [122]: 1 print("bid: {:.4f}, ask: {:.4f}, spread: {:.4f}".format(bid, ask, ask - bid))
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [123]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {ask - bid:.4f}")
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [124]: 1 print(f"bid: {bid:.4f}, ask: {ask:.4f}, spread: {(ask - bid):.4f}")
```

```
bid: 1.5760, ask: 1.5763, spread: 0.0003
```

```
In [125]: 1 string1 = "β, β, #"
          2 string1 = "Π, ó, θ, ω, v"
          3 chars = string1.split(", ")
          4 print(chars)
```

```
['Π', 'ó', 'θ', 'ω', 'v']
```

```
In [126]: 1 list1 = []
          2 for c in chars:
```

```
3 list1.append(hex(ord(c)))
4 print(list1)
```

```
['0x3a0', '0x3cd', '0x3b8', '0x3c9', '0x3bd']
```

```
In [127]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.upper())
          4 print(list1)
```

```
['Π', 'Υ', 'Θ', 'Ω', 'Ν']
```

```
In [128]: 1 list1 = []
          2 for c in chars:
          3     list1.append(c.lower())
          4 print(list1)
```

```
['π', 'ύ', 'θ', 'ω', 'ν']
```

```
In [129]: 1 a = 42
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 42 is even
```

```
In [130]: 1 a = 41
          2 print(f"The number {a} is not even") if a % 2 else print(f"The number {a} is even")
```

```
The number 41 is not even
```

```
In [131]: 1 a = 42
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 42 is even
```

```
In [132]: 1 a = 41
          2 print("The number {} is not even".format(a)) if a % 2 else print("The number {} is even".format(a))
```

```
The number 41 is not even
```

```
In [133]: 1 a = 0.3
          2 b = 0.1
          3 print(f"{a} / {b} = {a / b:.4f}")
```

```
0.3 / 0.1 = 3.0000
```

```
In [134]: 1 a = 42
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

```
The number 42 is even
```

```
In [135]: 1 a = 421
          2 qualifier = "is not" if a % 2 else "is"
          3 print(f"The number {a} {qualifier} even")
```

The number 421 is not even

```
In [136]: 1 print(range(10))
```

range(0, 10)

```
In [137]: 1 print(type(range(10)))
```

<class 'range'>

```
In [138]: 1 for i in range(3):
          2     print(i)
          3     print(i)
```

0
1
2
2

```
In [139]: 1 for i in "abc":
          2     pass
          3     print(i)
```

c

```
In [140]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
          2 for r in range(len(raggedArray)):
          3     for c in range(len(raggedArray[r])):
          4         print(f"[{r}, {c}] = {raggedArray[r][c]}")
          5     print("-" * 5)
```

[0, 0] = 1
[0, 1] = 2

[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6

[2, 0] = 7
[2, 1] = 8
[2, 2] = 9

[3, 0] = 10

```
In [141]: 1 identityMatrix = []
          2 size = 10
          3 for r in range(size):
          4     row = []
          5     for c in range(size):
          6         if r == c:
```

```

7         row.append(1)
8     else:
9         row.append(0)
10    identityMatrix.append(row)
11    identityMatrix

```

```

Out[141]: [[1, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0],
[0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
[0, 0, 0, 1, 0, 0, 0, 0, 0, 0],
[0, 0, 0, 0, 1, 0, 0, 0, 0, 0],
[0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
[0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
[0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 1, 0],
[0, 0, 0, 0, 0, 0, 0, 0, 0, 1]]

```

```

In [142]: 1 raggedArray = [[1, 2], [3, 4, 5, 6], [7, 8, 9], [10]]
2 for i, r in enumerate(raggedArray):
3     for j, e in enumerate(raggedArray[i]): # e for element, cuz it's not a
4         column technically
5             print(f"[{i}, {j}] = {raggedArray[i][j]}")
6             print("-" * 8)

```

```

[0, 0] = 1
[0, 1] = 2
-----
[1, 0] = 3
[1, 1] = 4
[1, 2] = 5
[1, 3] = 6
-----
[2, 0] = 7
[2, 1] = 8
[2, 2] = 9
-----
[3, 0] = 10
-----

```

```

In [143]: 1 print(f"{0.1:.4f}")

```

```
0.1000
```

```

In [144]: 1 print("{:.4f}".format(0.1))

```

```
0.1000
```

```

In [145]: 1 total, count = (0,) * 2
2 data = [10.5, 11.2, 9.8, None, 11.5, None]
3 for e in data:
4     if e:
5         total += e
6         count += 1
7 average = total / count
8 print(average)

```

```
10.75
```

```
In [146]: 1 for i, e in enumerate(data):
          2     if not e:
          3         data[i] = average
          4 print(data)
```

```
[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]
```

```
In [147]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
          2 count = sum(1 for e in data if e)
          3 total = sum(e for e in data if e)
          4 average = total / count
          5 print(sum, average)
```

```
<built-in function sum> 10.75
```

```
In [148]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
          2 average = statistics.fmean(e for e in data if e)
          3 data = [average if e is None else e for e in data]
          4 print(data)
```

```
[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]
```

```
In [149]: 1 data = [10.5, 11.2, 9.8, None, 11.5, None]
          2 df = pandas.DataFrame(data)
          3 df.fillna(df.mean(), inplace = True)
          4 print(df[0].tolist())
```

```
[10.5, 11.2, 9.8, 10.75, 11.5, 10.75]
```

```
In [150]: 1 print(len(str((1, 2))))
```

```
6
```

```
In [151]: 1 print(str((1, 2)))
```

```
(1, 2)
```

```
In [152]: 1 m, n = 4, 5
          2 for i in range(1, m):
          3     for j in range(1, n):
          4         print(f"{i} x {j} = {i * j}")
          5     print("-" * 8)
```

```
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
-----
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
-----
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
-----
```

In [153]:

```
1
2
3 data = ("2021-01-01", 10, 20], ["2021-01-02", 20, 18], ["2021-01-03", -10,
4 10], ["2021-01-04", 100, 102], ["2021-01-05", 20, 45])
5 largest = 0
6 for e in data:
7     distance = abs(e[2] - e[1])
8     e.append(distance)
9     if distance > largest:
10         dateOfLargest = e[0]
11         largest = distance
12 print(data)
13 print(dateOfLargest)
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```

```
(['2021-01-01', 10, 20, 10], ['2021-01-02', 20, 18, 2], ['2021-01-03', -10, 10, 20], ['2021-01-04',
100, 102, 2], ['2021-01-05', 20, 45, 25])
2021-01-05
```

In [154]:

```
1 data = [[10, 20], [20, 30], [35, 50], [45, 60]]
2 for i, r in enumerate(data):
3     if i == 0:
4         r.append("")
5     else:
6         if r[0] < data[i - 1][1]:
7             r.append("down")
8         elif r[0] > data[i - 1][1]:
9             r.append("up")
10        else:
11            r.append("same")
12 print(data)
```

```
[[10, 20, ''], [20, 30, 'same'], [35, 50, 'up'], [45, 60, 'down']]
```

```
In [155]: 1 tuple1 = ([0],)
          2 print(type(tuple1))
          3 print(tuple1)
          4 print(id(tuple1))
          5 print(id(tuple1[0]))
          6 tuple1[0][0] += 1
          7 print(tuple1)
          8 print(id(tuple1))
          9 print(id(tuple1[0]))
```

```
<class 'tuple'>
([0],)
128327770772736
128326959140992
([1],)
128327770772736
128326959140992
```

```
In [156]: 1 print(hash(True))
          2 a = True
          3 print(hash(a))
          4 b = True
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
1
1
1
1
```

```
In [157]: 1 print(hash(257))
          2 a = 257
          3 print(hash(a))
          4 b = 257
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
257
257
257
257
```

```
In [158]: 1 print(hash(2570))
          2 a = 2570
          3 print(hash(a))
          4 b = 2570
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
2570
2570
2570
2570
```

```
In [159]: 1 print(hash(25))
          2 a = 25
          3 print(hash(a))
          4 b = 25
          5 print(hash(b))
          6 a = b
          7 print(hash(a))
```

```
25
25
25
25
```

```
In [160]: 1 print(hash(2.5))
          2 a = 2.5
          3 print(hash(a))
          4 b = 2.5
          5 print(hash(b))
          6 print(hex(hash(b)))
          7 a = b
          8 print(len(hex(hash(a))))
          9 print(hash(b))
```

```
1152921504606846978
1152921504606846978
1152921504606846978
0x10000000000000002
18
1152921504606846978
```

```
In [161]: 1 for i in range(5):
          2     for j in range(4):
          3         pass
          4 print(i, j)
```

```
4 3
```

```
In [162]: 1 s1 = """
          2 "'And' and 'or' are the basic operations of logic. Together with 'no' (the
          3 logical operation
          4 of negation) they are a complete set of basic logical operations – all other
          5 logical
          6 operations, no matter how complex, can be obtained by suitable combinations of
          7 these."
          8 — John von Neumann, The Computer and the Brain
          9 """
          10
          11 s2 = """
          12 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor
          13 incididunt ut
```

```

7 labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
exercitation ullamco laboris
8 nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit
9 esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat
1 non proident, sunt
0 in culpa qui officia deserunt mollit anim id est laborum.
1 """
1
1 charCounts = {}
2 for c in s1:
1     if (ord(c) >= ord("A") and ord(c) <= ord("Z")) or (ord(c) >= ord("a")
3 and ord(c) <= ord("z")):
1         charCounts[c] = charCounts.get(c, 0) + 1
4 print(charCounts)
1
5
1
6
1
7
1
8
1
9
2
0

```

```

{'A': 1, 'n': 20, 'd': 4, 'a': 23, 'o': 31, 'r': 12, 'e': 27, 't': 21, 'h': 11, 'b': 7, 's': 9, 'i':
17, 'c': 10, 'p': 7, 'f': 4, 'l': 12, 'g': 6, 'T': 2, 'w': 2, 'y': 2, 'm': 6, 'x': 1, 'u': 3, 'J':
1, 'v': 1, 'N': 1, 'C': 1, 'B': 1}

```

```

In [163]: 1 d1 = {"a": 10, "b": 20, "c": 30}
2 d2 = {"d": 100, "e": 200, "f": 300}
3 d3 = {"f": 30, "g": 40}
4 keysList, valuesList = [], []
5 for d in d1, d2, d3:
6     for k, v in d.items():
7         keysList.append(k)
8         valuesList.append(v)
9 print(keysList)
10 print(valuesList)

```

```

['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]

```

```

In [164]: 1 keysList, valuesList = [], []
2 keysList = list(d1.keys())
3 keysList.extend(list(d2.keys()))
4 keysList.extend(list(d3.keys()))
5 print(keysList)

```

```

['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']

```

```
In [165]: 1 valuesList = list(d1.values())
          2 valuesList.extend(list(d2.values()))
          3 valuesList.extend(list(d3.values()))
          4 print(valuesList)
```

```
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [166]: 1 print(list(d1.keys()) + list(d2.keys()) + list(d3.keys()))
          2 print(list(d1.values()) + list(d2.values()) + list(d3.values()))
```

```
['a', 'b', 'c', 'd', 'e', 'f', 'f', 'g']
[10, 20, 30, 100, 200, 300, 30, 40]
```

```
In [167]: 1 grades = {"John": [90, 95, 98], "Eric": [86, 84, 92], "Michael": [90, 89, 85]}
          2 exam = {"Eric": 99, "John": 100}
          3 for k, v in grades.items():
          4     if k in exam:
          5         v.insert(0, exam[k])
          6     else:
          7         v.insert(0, None)
          8 print(grades)
```

```
{'John': [100, 90, 95, 98], 'Eric': [99, 86, 84, 92], 'Michael': [None, 90, 89, 85]}
```

```
In [168]: 1 for k, v in grades.items():
          2     score = exam.get(k)
          3     grades[k] = [score] + grades[k]
          4 print(grades)
```

```
{'John': [100, 100, 90, 95, 98], 'Eric': [99, 99, 86, 84, 92], 'Michael': [None, None, 90, 89, 85]}
```

```
In [169]: 1 for k, v in grades.items():
          2     v.insert(0, exam.get(k))
          3 print(grades)
```

```
{'John': [100, 100, 100, 90, 95, 98], 'Eric': [99, 99, 99, 86, 84, 92], 'Michael': [None, None, None, 90, 89, 85]}
```

```
In [170]: 1 for k in grades:
          2     print(k)
```

```
John
Eric
Michael
```

```
In [171]: 1 l = ["AAPL", "AAPL", "Aapl", "aapl", "MSFT"]
          2 print(list(set(l)))
```

```
['MSFT', 'aapl', 'AAPL', 'Aapl']
```

```
In [172]: 1 l = [e.casefold() for e in l]
          2 print(list(set(l)))
```

```
['aapl', 'msft']
```

```
In [173]: 1 se = {e.casefold() for e in l}
          2 print(se)
```

```
{'aapl', 'msft'}
```

```
In [174]: 1 data = {"d1": {"a": 1, "b": 2, "c": 3}, "d2": {"b": 20, "c": 30, "d": 40},
          2 "d3": {"d": 100, "x": 200}}
          3 uniqueKeys = {k for v in (data["d1"], data["d2"], data["d3"]) for k in v}
          4 print(uniqueKeys)
```

```
{'x', 'a', 'c', 'b', 'd'}
```

```
In [175]: 1 try:
          2     s1 = {[0, 1]}
          3     print(s1)
          4 except TypeError as ex:
          5     print(type(ex)., ":", " ", *ex.args, ".", sep = " ")
```

```
TypeError: cannot use 'list' as a set element (unhashable type: 'list').
```

```
In [176]: 1 m = [0] * 3
          2 print(m)
          3 print(m[0] is m[1])
```

```
[0, 0, 0]
True
```

```
In [177]: 1 m = [[0] * 3 for _ in range(3)]
          2 print(m)
          3 print(m[0] is m[1])
          4 print(m[0][0] is m[0][1])
```

```
[[0, 0, 0], [0, 0, 0], [0, 0, 0]]
False
True
```

```
In [178]: 1 m = [[10000 for i in range(3)] for _ in range(4)]
          2 print(m)
```

```
[[10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000], [10000, 10000, 10000]]
```

```
In [179]: 1 pprint.pprint(m, width = 1)
```

```
[[10000,
 10000,
 10000],
 [10000,
 10000,
 10000],
 [10000,
 10000,
 10000],
 [10000,
 10000,
 10000]]
```

```
In [180]: 1 print(m[0] is m[1])
          2 print(m[0][0] is m[0][1])
          3 print(hash(m[0][0]), hash(m[0][1]))
```

```
False
True
10000 10000
```

```
In [181]: 1 print(id(m[0][0]), id(m[0][1]))
```

```
128326959128176 128326959128176
```

```
In [182]: 1 m[0][0] = 1001
          2 pprint.pprint(m)
```

```
[[1001, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000],
 [10000, 10000, 10000]]
```

```
In [183]: 1 print(id(m[0][0]), id(m[0][1]))
          2 print(m[0][0] is m[0][1])
```

```
128326959126160 128326959128176
False
```

```
In [184]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = []
4 for d in data:
5     ranges.append(d["high"] - d["low"])
print(ranges)
```

```
[30, 50, 70, 90]
```

```
In [185]: data = [{"open": 100, "high": 120, "low": 90, "close": 110}, {"open": 110,
1 "high": 130, "low": 80, "close": 120}, {"open": 120, "high": 140, "low": 70,
2 "close": 130}, {"open": 130, "high": 150, "low": 60, "close": 140},]
3 ranges = [d["high"] - d["low"] for d in data]
print(ranges)
```

```
[30, 50, 70, 90]
```

```
In [186]: 1 bogusComposites = {i for i in range(1, 101) for j in range(2, 10) if i % j ==
          2 0}
          3 allNumbers = {i for i in range(11, 101)} | {1}
          print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

```
In [187]: 1 allNumbers = set(range(1, 101))
          2 print(allNumbers - bogusComposites)
```

```
{1, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}
```

In [188]:

```
data = [{"symbol": "ABCD", "name": "ABCD Company", "ranking": 2, "risk": 0.2},
{"symbol": "BCDE", "name": "BCDE Company", "ranking": 5, "risk": 0.2},
{"symbol": "CDEF", "name": "CDEF Company", "ranking": 8, "risk": 0.5},
{"symbol": "DEFG", "name": "DEFG Company", "ranking": 7, "risk": 0.8},
{"symbol": "EFGH", "name": "EFGH Company", "ranking": 9, "risk": 0.6},
{"symbol": "FGHI", "name": "FGHI Company", "ranking": 10, "risk": 0.1},
{"symbol": "GHIJ", "name": "GHIJ Company", "ranking": 3, "risk": 0.6},
{"symbol": "HIJK", "name": "HIJK Company", "ranking": 5, "risk": 0.5},
{"symbol": "IJKL", "name": "IJKL Company", "ranking": 5, "risk": 0.7},
{"symbol": "JKLM", "name": "JKLM Company", "ranking": 6, "risk": 0.9},
{"symbol": "KLMN", "name": "KLMN Company", "ranking": 6, "risk": 0.4},
{"symbol": "LMNO", "name": "LMNO Company", "ranking": 8, "risk": 0.4},
{"symbol": "MNOP", "name": "MNOP Company", "ranking": 8, "risk": 0.2},
{"symbol": "NOPQ", "name": "NOPQ Company", "ranking": 1, "risk": 0.5},
{"symbol": "OPQR", "name": "OPQR Company", "ranking": 9, "risk": 0.2},
{"symbol": "PQRS", "name": "PQRS Company", "ranking": 10, "risk": 0.9},
{"symbol": "QRST", "name": "QRST Company", "ranking": 3, "risk": 0.4},
{"symbol": "RSTU", "name": "RSTU Company", "ranking": 7, "risk": 0.3},
{"symbol": "STUV", "name": "STUV Company", "ranking": 8, "risk": 0.1},
{"symbol": "TUVW", "name": "TUVW Company", "ranking": 7, "risk": 0.9}]
filtered = [d for d in data if d["ranking"] > 5 and d["risk"] < 0.6]
print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1}]
```

In [189]:

```
1 for d in filtered:
2     d["weight"] = float(f"{d["ranking"] * d["risk"]:.2f}")
3 print(filtered)
```

```
[{'symbol': 'CDEF', 'name': 'CDEF Company', 'ranking': 8, 'risk': 0.5, 'weight': 4.0}, {'symbol': 'FGHI', 'name': 'FGHI Company', 'ranking': 10, 'risk': 0.1, 'weight': 1.0}, {'symbol': 'KLMN', 'name': 'KLMN Company', 'ranking': 6, 'risk': 0.4, 'weight': 2.4}, {'symbol': 'LMNO', 'name': 'LMNO Company', 'ranking': 8, 'risk': 0.4, 'weight': 3.2}, {'symbol': 'MNOP', 'name': 'MNOP Company', 'ranking': 8, 'risk': 0.2, 'weight': 1.6}, {'symbol': 'OPQR', 'name': 'OPQR Company', 'ranking': 9, 'risk': 0.2, 'weight': 1.8}, {'symbol': 'RSTU', 'name': 'RSTU Company', 'ranking': 7, 'risk': 0.3, 'weight': 2.1}, {'symbol': 'STUV', 'name': 'STUV Company', 'ranking': 8, 'risk': 0.1, 'weight': 0.8}]
```

In [190]:

```
1 weightedStocks = [dict(symbol = d["symbol"], weight = d["weight"]) for d in
2 filtered]
3 print(weightedStocks)
```

```
[{'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

In [191]:

```
1 weightedStocks = [dict(symbol = d["symbol"], weight = round(d["ranking"] *
2 d["risk"], 2)) for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]
3 print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [192]: 1 weightedStocks = [{"symbol": d["symbol"], "weight": round(d["ranking"] *  
2 d["risk"], 2)} for d in data if d["ranking"] >= 5 and d["risk"] < 0.6]  
print(weightedStocks)
```

```
[{'symbol': 'BCDE', 'weight': 1.0}, {'symbol': 'CDEF', 'weight': 4.0}, {'symbol': 'FGHI', 'weight': 1.0}, {'symbol': 'HIJK', 'weight': 2.5}, {'symbol': 'KLMN', 'weight': 2.4}, {'symbol': 'LMNO', 'weight': 3.2}, {'symbol': 'MNOP', 'weight': 1.6}, {'symbol': 'OPQR', 'weight': 1.8}, {'symbol': 'RSTU', 'weight': 2.1}, {'symbol': 'STUV', 'weight': 0.8}]
```

```
In [193]: 1 m = [[0] * 2] * 3  
2 print(m)  
3 for r in range(len(m)):  
4     for c in range(len(m[r])):  
5         print(id(m[r][c]))
```

```
[[0, 0], [0, 0], [0, 0]]  
11370000  
11370000  
11370000  
11370000  
11370000  
11370000
```

```
In [194]: 1 m = [[{0}] * 2] * 3  
2 print(m)  
3 for r in range(len(m)):  
4     for c in range(len(m[r])):  
5         print(id(m[r][c]))
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]  
128326958960640  
128326958960640  
128326958960640  
128326958960640  
128326958960640  
128326958960640
```

```
In [195]: 1 m[0][0] = 1  
2 for r in range(len(m)):  
3     for c in range(len(m[r])):  
4         print(id(m[r][c]))  
5 print(m)
```

```
11370032  
128326958960640  
11370032  
128326958960640  
11370032  
128326958960640  
[[1, {0}], [1, {0}], [1, {0}]]
```

```
In [196]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
          6 m[0][0] = 1
          7 print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
128326958957280
128326958960864
128326958959296
128326958957728
128326958961312
128326958959968
[[1, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [197]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
          2 print(m)
          3 for r in range(len(m)):
          4     for c in range(len(m[r])):
          5         print(id(m[r][c]))
          6 m[0][0] = {1}
          7 print(m)
```

```
[[{0}, {0}], [{0}, {0}], [{0}, {0}]]
128326958957280
128326958958400
128326958961984
128326958960192
128326958961088
128326958962656
[[{1}, {0}], [{0}, {0}], [{0}, {0}]]
```

```
In [198]: 1 m = [[0 for _ in range(2)] for _ in range(3)]
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
```

```
11370000
11370000
11370000
11370000
11370000
11370000
```

```
In [199]: 1 m = [[10000 for _ in range(2)] for _ in range(3)]
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
          5 m[0][0] = 1
          6 print(m)
```

```
128326959128624
128326959128624
128326959128624
128326959128624
128326959128624
128326959128624
[[1, 10000], [10000, 10000], [10000, 10000]]
```

```
In [200]: 1 m = [[{0} for _ in range(2)] for _ in range(3)]
          2 for r in range(len(m)):
          3     for c in range(len(m[r])):
          4         print(id(m[r][c]))
```

```
128326958958400
128326958961984
128326958960640
128326958957728
128326958962656
128326958961536
```

```
In [201]: 1 # print(input("Text box displayed ?"))
```

```
In [202]: 1 a = 0
          2 b = complex(real = 3.0, imag = 4)
          3 print(repr(a))
          4 print(repr(b))
          5 print(b)
```

```
0
(3+4j)
(3+4j)
```

```
In [203]: 1 values = [4, 3, 1, 2]
          2 try:
          3     minimum = abs(values[0])
          4     for value in values[1:]:
          5         if abs(value) < minimum:
          6             minimum = abs(value)
          7 except IndexError:
          8     minimum = 0
          9     pass
         10 finally:
         11     print(f"Minimum is: {minimum}")
```

```
Minimum is: 1
```

```
In [204]: 1 values = []
          2
          3 try:
          4     minimum = abs(values[0])
          5     for value in values[1:]:
          6         if abs(value) < minimum:
          7             minimum = abs(value)
          8 except IndexError:
          9     minimum = 0
```

```
10     pass
11 finally:
12     print(f"Minimum is: {minimum}")
```

Minimum is: 0

```
In [205]: 1 try:
2         raise ValueError("Custom message")
3 except ValueError as ex:
4         print(type(ex).      + ": " + ex.args[0] + ".")
```

ValueError: Custom message.

```
In [206]: 1 l = [4, 3, 2, 1, 0]
2         iterator = iter(l)
3         try:
4             while True:
5                 print(next(iterator))
6 except StopIteration as ex:
7         print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
4
3
2
1
0
StopIteration: .
```

```
In [207]: 1 try:
2         raise IndexError
3 except IndexError as ex:
4         print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

IndexError: .

```
In [208]: 1 rangeObject = range(10)
2         rangeIterator = iter(rangeObject)
3         print(next(rangeIterator))
4         print(next(rangeIterator))
5         print(next(rangeIterator))
6         print(list(rangeIterator))
```

```
0
1
2
[3, 4, 5, 6, 7, 8, 9]
```

```
In [209]: 1 rangeObj = range(5)
2         print(list(rangeObj))
3         print(list(rangeObj))
```

```
[0, 1, 2, 3, 4]
[0, 1, 2, 3, 4]
```

```
In [210]: 1 enumIterator = enumerate(string.ascii_lowercase)
2         print(enumIterator)
```

```

3 l = list(enumIterator)
4 print(l)
5 l = list(enumIterator)
6 print(dir(type(iter([0]))))
7 print(l)
8 try:
9     print(next(enumIterator))
10 except StopIteration as ex:
11     print(type(ex).      , ": ", *ex.args, ".", sep = "")

```

```

<enumerate object at 0x74b672ce8680>
[(0, 'a'), (1, 'b'), (2, 'c'), (3, 'd'), (4, 'e'), (5, 'f'), (6, 'g'), (7, 'h'), (8, 'i'), (9, 'j'),
(10, 'k'), (11, 'l'), (12, 'm'), (13, 'n'), (14, 'o'), (15, 'p'), (16, 'q'), (17, 'r'), (18, 's'),
(19, 't'), (20, 'u'), (21, 'v'), (22, 'w'), (23, 'x'), (24, 'y'), (25, 'z')]
['_class_', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__', '__ge__',
'__getattr__', '__getstate__', '__gt__', '__hash__', '__init__', '__init_subclass__',
'__iter__', '__le__', '__length_hint__', '__lt__', '__ne__', '__new__', '__next__', '__reduce__',
'__reduce_ex__', '__repr__', '__setattr__', '__setstate__', '__sizeof__', '__str__',
'__subclasshook__']
[]
StopIteration: .

```

```

In [211]: 1 print(type(enumIterator))
          2 print(type(iter(enumIterator)))
          3 print(issubclass(type(iter([0])), type(enumIterator)))

```

```

<class 'enumerate'>
<class 'enumerate'>
False

```

```

In [212]: 1 print(iter(enumIterator) is enumIterator)

True

```

```

In [213]: 1 print(type(iter({0, 1, 2})))

<class 'set_iterator'>

```

```

In [214]: 1 print(type(iter({0: "0", 1: "1", "2": 2})))

<class 'dict_keyiterator'>

```

```

In [215]: 1 l = [4, 3, 2, 1, 0]
          2 iterator = iter(l)
          3 for i in iterator:
          4     print(i)

```

```

4
3
2
1
0

```

```

In [216]: 1 try:
          2     print(next(l))
          3 except TypeError as ex:
          4     print(type(ex).      , ": ", *ex.args, ".", sep = "")

```

TypeError: 'list' object is not an iterator.

```
In [217]: 1 start = time.perf_counter()
          2 squaresGenerator = (i ** 10 for i in range(1_000_000))
          3 end = time.perf_counter()
          4 print(f"Elapsed: {end - start}")
```

Elapsed: 0.00010408199159428477

```
In [218]: 1 start = time.perf_counter()
          2 squaresListComp = [i ** 10 for i in range(1_000_000)]
          3 end = time.perf_counter()
          4 print(f"Elapsed: {end - start}")
```

Elapsed: 0.20623250800417736

```
In [219]: 1 print((7.16e-5 / 1.824e-1) ** -1)
          2 print(105 / 9)
          3 print(10.5e1 / 9e0)
          4 print(99 / 1)
          5 print(9.9e1 / 1e0) # 1 order of magnitude difference means a multiple of
                                between 10 and 100
```

2547.486033519553
11.666666666666666
11.666666666666666
99.0
99.0

```
In [220]: 1 print(999_999 ** 10)
```

99999000000449998800002099997480002099998800000044999990000001

```
In [221]: 1 print(squaresGenerator is iter(squaresGenerator))
```

True

```
In [222]: 1 l = [10, "abc", 3.14, True]
          2 for i, e in enumerate(l):
          3     print(f"{i}: {e}")
```

0: 10
1: abc
2: 3.14
3: True

```
In [223]: 1 def f(a, *b, x, **y):
          2     print("Positional/Named:", a)
          3     print("Variable positional:", b)
          4     print("Named only:", x)
          5     print("Variable named:", y)
          6
          7
          8 f(0, 1, 2, x = 3, y = 4, z = 5)
```

Positional/Named: 0
Variable positional: (1, 2)
Named only: 3
Variable named: {'y': 4, 'z': 5}

```
In [224]: 1 # f(10, 2, x = 10, 20)
```

```
In [225]: 1 f(10, 2, x = 10, y = 20)
```

Positional/Named: 10
Variable positional: (2,)
Named only: 10
Variable named: {'y': 20}

```
In [226]: 1 f(10, z = 2, x = 10, y = 20)
```

Positional/Named: 10
Variable positional: ()
Named only: 10
Variable named: {'z': 2, 'y': 20}

```
In [227]: 1 f(10, x = 2, y = 10, z = 20)
```

Positional/Named: 10
Variable positional: ()
Named only: 2
Variable named: {'y': 10, 'z': 20}

```
In [228]: 1 # f(a = 10, 2, x = 10, y = 20)
```

```
In [229]: 1 f(x = 10, y = 20, z = 30, a = 10)
```

Positional/Named: 10
Variable positional: ()
Named only: 10
Variable named: {'y': 20, 'z': 30}

```
In [230]: 1 try:
2         f(10, a = 2, x = 10, y = 20)
3     except TypeError as ex:
4         print(type(ex).      )
```

TypeError

```
In [231]: 1 def f1(a, *, x, **y):
2         print(f"a: {a}, x: {x}, y: {y}")
```

```
In [232]: 1 try:
2         f1(10, a = 20, x = 2, y = 10)
3     except TypeError as ex:
4         print(type(ex).      )
```

TypeError

```
In [233]: 1 f1(a = 1, x = 2)
```

a: 1, x: 2, y: {}

```
In [234]: 1 print(max("cba"))
```

c

```
In [235]: 1 def f2(a, b):
2         print(f"a: {a}; b: {b}")
3
4
5 f2(2, 3)
```

a: 2; b: 3

```
In [236]: 1 f2(2, b = 3)
```

a: 2; b: 3

```
In [237]: 1 # f2(a = 2, 3)
```

```
In [238]: 1 f2(b = 1, a = 0)
```

a: 0; b: 1

```
In [239]: 1 # def f3(a = 0, b):
2         #     print(f"a: {a}; b: {b}")
```

```
In [240]: 1 def f3(a, b = 0):
2         print(f"a: {a}; b: {b}")
3
4
5 f3(-1, 1)
```

a: -1; b: 1

```
In [241]: 1 # f3(a = -1, 1)
```

```
In [242]: 1 f3(a = -1, b = 1)
```

a: -1; b: 1

```
In [243]: 1 def f4(a, b = 1, *, x):
2         print(a, b, x)
3
4
5 f4(0, 2, x = 3)
```

0 2 3

```
In [244]: 1 def f5(a, b = 1, *, x = 2, y):
2         print(a, b, x, y)
3
4
5 f5(0, 2, x = 3, y = 4)
```

0 2 3 4

```
In [245]: 1 def f6(**x):
          2     print(x)
          3
          4
          5 f6(a = 1, b = 2, c = 3)
```

{'a': 1, 'b': 2, 'c': 3}

```
In [246]: 1 # def f7(*, **x):
          2 #     print(x)
          3 #
          4 #
          5 # f7(a = 1, b = 2)
```

```
In [247]: 1 def f8(a, *, b):
          2     print(f"a: {a}; b: {b}")
          3
          4
          5 f8(b = 1, a = 0)
```

a: 0; b: 1

```
In [248]: 1 def toJSON(a, *, b, **c):
          2     print(a, b, c)
          3
          4
          5 toJSON(0, b = 3, c1 = 0, c2 = 1)
```

0 3 {'c1': 0, 'c2': 1}

```
In [249]: 1 def avg(a, *b):
          2     return (sum(b) + a) / (len(b) + 1)
          3
          4
          5 print(avg(0, 1, 2, 3, 4))
          6 print(avg(0))
          7 print(avg(0, 1))
```

2.0
0.0
0.5

```
In [250]: 1 def separator(b = 10, *, a = "-"):
          2     return a * b
          3
          4
          5 print(separator())
          6 print(separator(a = "A"))
          7 print(separator(5))
          8 print(separator(a = "B", b = 5))
          9 print(separator(5, a = "B"))
```

```
-----  
AAAAAAAAA  
-----  
BBBBBB  
BBBBBB
```

```
In [251]: 1 print(sep = "|", *("AB", "CD"))
```

```
AB|CD
```

```
In [252]: 1 # def f(a, *b, c):  
2         #     print(a, b, c)  
3         #  
4         #  
5         # f(c = 2, 0, 1, 2)
```

```
In [253]: 1 # def f(*a, b = 0):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0, 2)
```

```
In [254]: 1 try:  
2         f(b = 1, *(0, 2))  
3 except TypeError as ex:  
4         print(type(ex).      )
```

```
TypeError
```

```
In [255]: 1 # def f(a, *, b):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [256]: 1 # def f(a, b):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [257]: 1 # def f(a, b = 2):  
2         #     print(a, b)  
3         #  
4         #  
5         # f(b = 1, 0)
```

```
In [258]: 1 def f(a, b = 2):  
2         print(a, b)  
3
```

```
4
5 f(b = 1, *(0,))
```

```
0 1
```

```
In [259]: 1 anonymous = lambda iterable: len(set(iterable))
2          print(anonymous("abcd"))
3          print(anonymous("abab"))
4          try:
5              print(anonymous())
6          except TypeError as ex:
7              print(type(ex).      )
8          print(anonymous(""))
9          print(anonymous([0, 1, 2]))
10         print(anonymous([0]))
11         print(anonymous([]))
12         print(anonymous([1, 1, 2, 2, 3, 4]))
```

```
4
2
TypeError
0
3
1
0
4
```

```
In [260]: 1 s = "This is the first sentence. This is the scecond sentence. This is not the
2          fourth sentence, it is the third sentence."
3
4          def f(s = ""):
5              return dict(collections.Counter(s))
6
7
8          print(f(s))
9          print(f(""))
```

```
{'T': 3, 'h': 9, 'i': 10, 's': 13, ' ': 20, 't': 13, 'e': 17, 'f': 2, 'r': 3, 'n': 10, 'c': 6, 'l':
3, 'o': 3, 'd': 2, 'u': 1, ',': 1}
{}
```

```
In [261]: 1 s = "This is the first sentence. This is the scecond sentence. This is not the
2          fourth sentence, it is the third sentence."
3
4          def f(s = ""):
5              words = re.split("\\.| |", s)
6              for word in words:
7                  if word == "":
8                      words.remove(word)
9              frequencies = dict.fromkeys(set(words), 0)
10             for word in words:
11                 frequencies[word] += 1
12             print(frequencies)
```

```
7
8 f(s)
9
10
11
12
13
14
15
```

```
{'the': 4, 'is': 4, 'sentence': 4, 'first': 1, 'scecond': 1, 'fourth': 1, 'This': 3, 'not': 1, 'third': 1, 'it': 1}
```

```
In [262]: 1 def f(s = ""):
2         s = s.replace(".", "")
3         s = s.replace(",", "")
4         words = s.split()
5         return dict(collections.Counter(words))
6
7
8 print(f(s))
```

```
{'This': 3, 'is': 4, 'the': 4, 'first': 1, 'sentence': 4, 'scecond': 1, 'not': 1, 'fourth': 1, 'it': 1, 'third': 1}
```

```
In [263]: 1 print(round(1.812))
```

```
2
```

```
In [264]: 1 n, d = (0.3).as_integer_ratio()
2         print(n, d)
3         print(1 / 3)
```

```
5404319552844595 18014398509481984
0.3333333333333333
```

```
In [265]: 1 print(round(0.325, 2))
```

```
0.33
```

```
In [266]: 1 print(f"{0.325:.40f}")
```

```
0.3250000000000000111022302462515654042363
```

```
In [267]: 1 def toJSON(a, *, b, **c):
2         print(a, b, c)
3
```

```

4
5 try:
6     toJSONN(0, 1, 2, b = 3, c1 = 4, c2 = 5)
7 except TypeError as ex:
8     print(type(ex).

```

TypeError

```

In [268]: 1 def toJSONN(a, *, b, **c):
2         print(a, b, c)
3
4
5 toJSONN(0, b = 3, c1 = 4, c2 = 5)

```

0 3 {'c1': 4, 'c2': 5}

```

In [269]: 1 # def toJSONN(a, *):
2         #     print(a)
3         #
4         #
5         # toJSONN(0, 1, 2)

```

```

In [270]: 1 def toJSONN(a, *args):
2         print(a, args)
3
4
5 toJSONN(0, 1, 2)

```

0 (1, 2)
(1, 2)

```

In [271]: 1 l = ["AB CD", "EF CC", "GH BC", "IJ AC", "KL CA", "MN JQ", "OP BA"]
2         print(sorted(l, key = lambda e: (e[3], e[4])))

```

['IJ AC', 'OP BA', 'GH BC', 'KL CA', 'EF CC', 'AB CD', 'MN JQ']

```

In [272]: 1 # print(sorted(l, key = lambda e: e[len(e) - i] for i in range(len(e))))
2         print(sorted(l, key = lambda e: e[len(e) - 1]))

```

['KL CA', 'OP BA', 'EF CC', 'GH BC', 'IJ AC', 'AB CD', 'MN JQ']

```

In [273]: 1 a = (1 for _ in range(5))
2         print(a)
3         print(lambda _: "ABCD")
4         print(sorted)
5         print(range)

```

<generator object <genexpr> at 0x74b672d1e810>
<function <lambda> at 0x74b672cc7480>
<built-in function sorted>
<class 'range'>

```

In [274]: 1 def f(*a, b):
2         print(a, b)
3

```

```
4
5 f(b = 0, *(1, 2))
```

```
(1, 2) 0
```

```
In [275]: 1 def f(a, *b, c):
          2     print(a, b, c)
          3
          4
          5 f(0, c = 1, *(2, 3))
```

```
0 (2, 3) 1
```

```
In [276]: 1 print(statistics.fmean)
```

```
<function fmean at 0x74b6a3cf2980>
```

```
In [277]: 1 print(dict([("a", 97), ("b", 98), ("c", 99)]))
```

```
{'a': 97, 'b': 98, 'c': 99}
```

```
In [278]: 1 widgets = [f"w{i}" for i in range(1, 21)]
          2 skus = [f"sku{i}" for i in range(1, len(widgets) + 1)]
          3 print(widgets)
          4 print(skus)
```

```
['w1', 'w2', 'w3', 'w4', 'w5', 'w6', 'w7', 'w8', 'w9', 'w10', 'w11', 'w12', 'w13', 'w14', 'w15',
'w16', 'w17', 'w18', 'w19', 'w20']
['sku1', 'sku2', 'sku3', 'sku4', 'sku5', 'sku6', 'sku7', 'sku8', 'sku9', 'sku10', 'sku11', 'sku12',
'sku13', 'sku14', 'sku15', 'sku16', 'sku17', 'sku18', 'sku19', 'sku20']
```

```
In [279]: 1 print({w: s for w, s in zip(widgets, skus)})
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
'sku20'}
```

```
In [280]: 1 print(dict(zip(widgets, skus)))
```

```
{'w1': 'sku1', 'w2': 'sku2', 'w3': 'sku3', 'w4': 'sku4', 'w5': 'sku5', 'w6': 'sku6', 'w7': 'sku7',
'w8': 'sku8', 'w9': 'sku9', 'w10': 'sku10', 'w11': 'sku11', 'w12': 'sku12', 'w13': 'sku13', 'w14':
'sku14', 'w15': 'sku15', 'w16': 'sku16', 'w17': 'sku17', 'w18': 'sku18', 'w19': 'sku19', 'w20':
'sku20'}
```

```
In [281]: 1 suits = "shdc" # Spades, Hearts, Diamonds, Clubs
          2 ranks = list("23456789") + ["10", "J", "Q", "K", "A"]
          3 print(suits)
          4 print(ranks)
```

```
shdc
['2', '3', '4', '5', '6', '7', '8', '9', '10', 'J', 'Q', 'K', 'A']
```

```
['2', '3', '4', '5', '6', '7', '8', '9', '10', 'J', 'Q', 'K', 'A']
```

```
In [282]: 1 print([(rank, suit) for rank in ranks] for suit in suits])
```

```
[('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'), ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's')], [('2', 'h'), ('3', 'h'), ('4', 'h'), ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'), ('K', 'h'), ('A', 'h')], [('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'), ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd')], [('2', 'c'), ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'), ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c')]]
```

```
In [283]: 1 pprint.pprint([(rank, suit) for rank in ranks] for suit in suits])
```

```
[('2', 's'),
 ('3', 's'),
 ('4', 's'),
 ('5', 's'),
 ('6', 's'),
 ('7', 's'),
 ('8', 's'),
 ('9', 's'),
 ('10', 's'),
 ('J', 's'),
 ('Q', 's'),
 ('K', 's'),
 ('A', 's')],
 [('2', 'h'),
 ('3', 'h'),
 ('4', 'h'),
 ('5', 'h'),
 ('6', 'h'),
 ('7', 'h'),
 ('8', 'h'),
 ('9', 'h'),
 ('10', 'h'),
 ('J', 'h'),
 ('Q', 'h'),
 ('K', 'h'),
 ('A', 'h')],
 [('2', 'd'),
 ('3', 'd'),
 ('4', 'd'),
 ('5', 'd'),
 ('6', 'd'),
 ('7', 'd'),
 ('8', 'd'),
 ('9', 'd'),
 ('10', 'd'),
 ('J', 'd'),
 ('Q', 'd'),
 ('K', 'd'),
 ('A', 'd')],
 [('2', 'c'),
 ('3', 'c'),
 ('4', 'c'),
 ('5', 'c'),
 ('6', 'c'),
 ('7', 'c'),
 ('8', 'c'),
 ('9', 'c'),
 ('10', 'c'),
 ('J', 'c'),
 ('Q', 'c'),
 ('K', 'c'),
 ('A', 'c')]]
```

```
In [284]: 1 print([list(zip(ranks, s * 13)) for s in suits])
```

```
[[('2', 's'), ('3', 's'), ('4', 's'), ('5', 's'), ('6', 's'), ('7', 's'), ('8', 's'), ('9', 's'), ('10', 's'), ('J', 's'), ('Q', 's'), ('K', 's'), ('A', 's')], [('2', 'h'), ('3', 'h'), ('4', 'h'), ('5', 'h'), ('6', 'h'), ('7', 'h'), ('8', 'h'), ('9', 'h'), ('10', 'h'), ('J', 'h'), ('Q', 'h'), ('K', 'h'), ('A', 'h')], [('2', 'd'), ('3', 'd'), ('4', 'd'), ('5', 'd'), ('6', 'd'), ('7', 'd'), ('8', 'd'), ('9', 'd'), ('10', 'd'), ('J', 'd'), ('Q', 'd'), ('K', 'd'), ('A', 'd')], [('2', 'c'), ('3', 'c'), ('4', 'c'), ('5', 'c'), ('6', 'c'), ('7', 'c'), ('8', 'c'), ('9', 'c'), ('10', 'c'), ('J', 'c'), ('Q', 'c'), ('K', 'c'), ('A', 'c')]]
```

In [285]:

```
1 def f(list1, *, reverse = False):
2     if not reverse:
3         return sorted(list1)
4     else:
5         return sorted(list1, reverse = True)
6
7
8 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
9 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
10 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
11 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

In [286]:

```
1 def f(list1, *, reverse = False):
2     return sorted(list1, reverse = reverse)
3
4
5 print(f([3, 2, 4, 5, 1, 6, 8, 7]))
6 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True))
7 print(f([3, 2, 4, 5, 1, 6, 8, 7])[0])
8 print(f([3, 2, 4, 5, 1, 6, 8, 7], reverse = True)[0])
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
[8, 7, 6, 5, 4, 3, 2, 1]
1
8
```

In [287]:

```
1 print(hex(ord(chr(0xE3).upper())))
```

0xc3

In [288]:

```
1 print("abc" in "defabcghi")
```

True

In [289]:

```
1 print("abc" in "abbde")
```

False

In [290]:

```
1 def solve(a, b, c):
2     solution = ((-1 * b) + math.sqrt((b * b) - (4 * a * c))) / (2 * a)
3     return round(solution, 2)
4
5
```

```
6 print(solve(3, 4, -5))
7 print(solve(1, -5, -8))
```

```
0.79
6.27
```

```
In [291]: 1 def encrypt(s):
          2     return "".join(chr(ord(c) + 10) for c in s)
          3
          4
          5 print(encrypt("ABCDEFGHIJKLMNOPQRSTUVWXYZ"))
```

```
KLMNOPQRSTUVWXYZ[\]^_`abcd
```

```
In [292]: 1 def decrypt(s):
          2     return "".join(chr(ord(c) - 10) for c in s)
          3
          4
          5 print(decrypt("KLMNOP"))
          6 print(decrypt("S}kkm* Xo\x81~yx"))
```

```
ABCDEF
Isaac Newton
```

```
In [293]: 1 print(10 % 3)
```

```
1
```

```
In [294]: 1 a = array.array("l", [20])
          2 print(a)
```

```
array('l', [20])
```

```
In [295]: 1 for i in a:
          2     print(i)
```

```
20
```

```
In [296]: 1 print(sys.getsizeof(a))
```

```
88
```

```
In [297]: 1 try:
          2     print(sys.getallocatedblocks(a))
          3 except TypeError as e:
          4     print(type(e).      )
```

```
TypeError
```

```
In [298]: 1 print(sys.getsizeof(1))
```

```
28
```

```
In [299]: 1 def f(s):
2         print(s)
3
4
5     try:
6         f(*"hello")
7     except TypeError as ex:
8         print(type(ex).      , ex.args)
```

TypeError ('f() takes 1 positional argument but 5 were given',)

```
In [300]: 1 def func(s):
2         print(s)
3
4
5     def apply(func, *str):
6         result = func(*str)
7         return result
8
9
10    apply(func, "john")
```

john

```
In [301]: 1 try:
2         apply(func, "john", "smith")
3     except TypeError as ex:
4         print(type(ex).      , *ex.args)
```

TypeError func() takes 1 positional argument but 2 were given

```
In [302]: 1 def encloseGetPower(b):
2         print("hex(id(b)):", hex(id(b)))
3
4         def getPower(e):
5             print("hex(id(e)):", hex(id(e)))
6
7             return b ** e
8
9         print("getPower:", getPower)
10
11        return getPower
12
13
14    getBinaryPower = encloseGetPower(2)
15    print(getBinaryPower(31))
16    getTertiaryPower = encloseGetPower(3)
17    print(getTertiaryPower(4))
```

```
hex(id(b)): 0xad7e50
getPower: <function encloseGetPower.<locals>.getPower at 0x74b672cc75e0>
hex(id(e)): 0xad81f0
2147483648
hex(id(b)): 0xad7e70
getPower: <function encloseGetPower.<locals>.getPower at 0x74b672cc7c10>
hex(id(e)): 0xad7e90
81
```

```
In [303]: 1 print("getBinaryPower:", getBinaryPower)
          2 print("encloseGetPower:", encloseGetPower)
          3 try:
          4     print("getPower:", getPower)
          5 except NameError as ex:
          6     print(type(ex).      , ex.args)
```

```
getBinaryPower: <function encloseGetPower.<locals>.getPower at 0x74b672cc75e0>
encloseGetPower: <function encloseGetPower at 0x74b672cc7110>
NameError ("name 'getPower' is not defined",)
```

```
In [304]: 1 print("getBinaryPower:", getBinaryPower.      )
          2 print("encloseGetPower:", encloseGetPower.      )
```

```
getBinaryPower: (<cell at 0x74b67208dc00: int object at 0xad7e50>,)
encloseGetPower: None
```

```
In [305]: 1 def outer(a, b):
          2     print(hex(id(a)))
          3     sumTotal = a + b
          4
          5     def inner():
          6         prod = a * b
          7         print(a, b, sumTotal, prod)
          8         return "You just called a closure!"
          9
         10     return inner
         11
         12
         13 f = outer(2, 3)
```

```
0xad7e50
```

```
In [306]: 1 f = outer(2, 3)
```

```
0xad7e50
```

```
In [307]: 1 print(f)
```

```
<function outer.<locals>.inner at 0x74b672cc6820>
```

```
In [308]: 1 print(f.      )
```

```
(<cell at 0x74b672d287c0: int object at 0xad7e50>, <cell at 0x74b67208fe80: int object at 0xad7e70>,
<cell at 0x74b66f1407f0: int object at 0xad7eb0>)
```

```
In [309]: 1 f()
```

2 3 5 6

Out[309]: 'You just called a closure!'

```
In [310]: 1 def execute(func):
2
3           def inner(a, b):
4               return func(a, b)
5
6           return inner
7
8
9 def add(a, b):
10     return a + b
11
12
13 add_executor = execute(add)
14 print(add_executor.)
```

(<cell at 0x74b66f140640: function object at 0x74b672cc78a0>,)

```
In [311]: 1 print(hex(id(add)))
```

0x74b672cc78a0

```
In [312]: 1 try:
2           print(hex(add))
3 except TypeError as ex:
4     print(type(ex).)
```

TypeError

```
In [313]: 1 add_executor(2, 3)
```

Out[313]: 5

```
In [314]: 1 def f():
2           pass
3
4
5 print(f.)
```

None

```
In [315]: 1 def f():
2
3           def g():
4               return
5
6           return g
7
8
9 h = f()
```

```
10 print(f.      )
11 print(h.      )
```

None
None

```
In [316]: 1 a = 0
          2 try:
          3     print(a.      )
          4 except AttributeError as e:
          5     print(type(e).      )
```

AttributeError

```
In [317]: 1 a = []
          2 try:
          3     print(a.      )
          4 except AttributeError as e:
          5     print(type(e).      )
```

AttributeError

```
In [318]: 1 def execute(func):
          2
          3     def inner(*posargs, **kwargs):
          4         return func(*posargs, **kwargs)
          5
          6     return inner
          7
          8
          9 def add(a, b, c):
          10     return a + b + c
          11
          12
          13 def say_hello(name, *, formal = True):
          14     return f"Pleased to meet you, {name}" if formal else f"Hi, {name}!"
          15
          16
          17 exec_add = execute(add)
          18 exec_greet = execute(say_hello)
```

```
In [319]: 1 exec_add(1, 2, 3)
```

Out[319]: 6

```
In [320]: 1 exec_greet("Michael", formal = False)
```

Out[320]: 'Hi, Michael!'

```
In [321]: 1 exec_greet("Michael", formal = True)
```

Out[321]: 'Pleased to meet you, Michael'

```
In [322]: 1 print((0.1).as_integer_ratio())
```

```
(3602879701896397, 36028797018963968)
```

```
In [323]: 1 print((0.125).as_integer_ratio())
```

```
(1, 8)
```

```
In [324]: 1 print((0.2).as_integer_ratio())
2 print(f"{0.2:.40f}")
```

```
(3602879701896397, 18014398509481984)
0.2000000000000000111022302462515654042363
```

```
In [325]: 1 print(math.pi.as_integer_ratio())
2 n, d = math.pi.as_integer_ratio()
3 print(n / d)
```

```
(884279719003555, 281474976710656)
3.141592653589793
```

```
In [326]: 1 try:
2         a1 = a1 + 1
3         print(a1)
4     except NameError as ex:
5         print(type(ex).      , ": ", *ex.args, ".", sep = "")
```

```
NameError: name 'a1' is not defined.
```

```
In [327]: 1 def getDiagonalMatrix(val, rows, columns):
2         diagonalMatrix = [[val if ce == r else 0 for ce in range(4)] for r in
3         range(5)]
4         return diagonalMatrix
```

```
In [328]: 1 def getDiagonalMatrix(rows, columns, *, val):
2         diagonalMatrix = [[val if cl == row else 0 for cl in range(columns)]
3         for row in range(rows)]
4         return diagonalMatrix
```

```
In [329]: 1 def getFactorial(n):
2         factorial = 1
3         for i in range(1, n + 1):
4             factorial *= i
5         return factorial
```

```
In [330]: 1 result = getFactorial(5)
2 print(result)
```

```
120
```

```
In [331]: 1 result = getDiagonalMatrix(3, 4, val = 5)
2 pprint.pprint(result, width = 14)
```

```
[[5, 0, 0, 0],  
[0, 5, 0, 0],  
[0, 0, 5, 0]]
```

```
In [332]: 1 print(f"{timeit.timeit('getFactorial(2)', globals = globals()):.4g}")
```

0.1301

```
In [333]: 1 n = 0.0000000000000123456789  
2 n = 4.482882e-05  
3 exp = 0  
4 while n < 1:  
5     exp -= 3  
6     n *= 1000  
7 prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f", "a"  
8 print(f"{n}{prefixes[abs(exp)//3]}")
```

44.82882μ

```
In [334]: 1 n = 10000000000  
2 exp = 0  
3 while n >= 1000:  
4     exp += 3  
5     n /= 1000  
6 prefixes = "", "k", "M", "G", "T", "P", "E"  
7 print(f"{n}{prefixes[exp//3]}")
```

1.0G

```
In [335]: def toEnggNotation(n, scientificNotation = True, nSignificantDigits = 4):  
1         exp = 0  
2         oldN = n  
3         if n > 1:  
4             while n >= 1000:  
5                 exp += 3  
6                 n /= 1000  
7                 prefixes = "", "k", "M", "G", "T", "P", "E"  
8                 return f"{n:.3f}{prefixes[exp//3]}\t{oldN}" if  
9 scientificNotation else f"{n}{prefixes[exp//3]}"  
10         elif n < 1:  
11             while n < 1:  
12                 exp -= 3  
13                 n *= 1000  
14                 prefixes = "", "m", "\N{GREEK SMALL LETTER MU}", "n", "p", "f",  
15 "a"  
16             return f"{n:.{nSignificantDigits}f}  
17 {prefixes[abs(exp)//3]}\t{oldN}" if scientificNotation else f"{n}  
18 {prefixes[abs(exp)//3]}"  
19         else:  
20             return 1
```



```

3         print(f(*pos, **keyword))
4         end = time.perf_counter()
5         return end - start
6
7
8 print(toEnggNotation(myTimer(getFactorial, 1000)))
9 print(toEnggNotation(myTimer(getDiagonalMatrix, 3, 5, val = 7)))

```

[illegible]

```

1 def encloseMyTimer(f):
2
3     def myTimer(*pos, **keyword):
4         start = time.perf_counter()
5         result = f(*pos, **keyword)
6         end = time.perf_counter()
7         print(toEnggNotation(end - start, True, 4))
8         return result
9
10    return myTimer
11
12
13 timed = encloseMyTimer(getDiagonalMatrix)
14 print(timed(2, 4, val = -5))
15 timed = encloseMyTimer(getFactorial)
16 print(timed(11))
17 print(encloseMyTimer(getDiagonalMatrix)(2, 4, val = -5))
18 print(encloseMyTimer(getFactorial)(11))

```

```
3.1760μ 3.176013706251979e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
1.7430μ 1.7429993022233248e-06
39916800
2.5340μ 2.5339832063764334e-06
[[-5, 0, 0, 0], [0, -5, 0, 0]]
1.1620μ 1.1619995348155499e-06
39916800
```

```
In [340]: 1 # print(eval("2 * 3"))
          2 # print(eval("x = 2 and print(x)"))
```

```
In [341]: 1 # eval("x = 2")
          2 # print(x)
```

```
In [342]: 1 x = 2
          2 eval("x ** 2 - 1")
```

```
Out[342]: 3
```

```
In [343]: 1 def f(expr, x):
          2     print(eval(expr))
          3
          4
          5 x = 2
          6 f("x ** 2 - 1", x)
```

```
3
```

```
In [344]: 1 def f(x):
          2     return True
          3
          4
          5 try:
          6     print(list(map(f(False), range(0, 5))))
          7 except TypeError as ex:
          8     print(type(ex).      + ":", ex.args[0])
```

```
TypeError: 'bool' object is not callable
```

```
In [345]: 1 def getExtremum(expr, interval, granularity, minimax):
          2
          3     def f(expr):
          4
          5         def g(x):
          6             return eval(expr)
          7
          8         return g
          9
          10     evalExpr = f(expr)
          11     print("Range:", list(range(interval[0], interval[1] + granularity,
          12 granularity)))
          13     if minimax == "minimum":
```

```

7         return min(map(evalExpr, range(interval[0], interval[1] +
granularity, granularity)))
8         elif minimax == "maximum":
9             return max(map(evalExpr, range(interval[0], interval[1] +
granularity, granularity)))
1        else:
10           return "Invalid minimax"
11
12
13
14
15
16
17
18
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20
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96
97
98
99

```

```

Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
-1
Range: [-10, -8, -6, -4, -2, 0, 2, 4, 6, 8, 10]
99

```

```

In [346]: 1 expr = "x ** 2 - 1"
          2 f = lambda x: eval(expr)
          3 print(f(3))

```

8

```

In [347]: 1 def getExtremum(expr, interval, resolution, minimax):
          2
          3     def f(expr):
          4         def g(x):
          5             return eval(expr)
          6
          7         return g
          8
          9     evalExpr = f(expr)

```

```

    l = [interval[0] + (i * (interval[1] - interval[0]) / (resolution -
6    1)) for i in range(resolution)] # Cheated with the calc.
        return min(map(evalExpr, l)) if minimax == "minimum" else
7    max(map(evalExpr, l))

8
    expr = "x ** 2 - 1"
9    print(getExtremum(expr, (-10, 10), 5, "minimum"))
1    print(getExtremum(expr, (-10, 10), 5, "maximum"))
0    print(getExtremum(expr, (-10, 10), 6, "minimum"))
1    print(getExtremum(expr, (-10, 10), 6, "maximum"))
1    expr = "math.log10(x)"
1    print(getExtremum(expr, (1, 100), 20, "minimum"))
2    print(getExtremum(expr, (1, 100), 20, "maximum"))
1    expr = "math.sin(x)"
3    print(getExtremum(expr, (-10, 10), 20, "minimum"))
1    print(getExtremum(expr, (-10, 10), 20000, "maximum"))
4    expr = "x"
1    print(getExtremum(expr, (1, 6), 8, "minimum"))
5    print(getExtremum(expr, (1, 6), 8, "maximum"))

1
6
1
7
1
8
1
9
2
0
2
1
2
2
2
3
2
4
2
5
2
6
2
7
2
8

```

```
-1.0
99.0
3.0
99.0
0.0
2.0
-0.9999667804441291
0.999999996044897
1.0
6.0
```

In [348]:

```
def f(t):
    return t[0] ** t[1]

l = [(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (3, 4), (4, 3),
      (4, 4)]

start = time.perf_counter()
for _ in range(10_000):
    result = []
    for t in l:
        result.append(f(t))
end = time.perf_counter()
forLoopTime = end - start
print(result)

start = time.perf_counter()
for _ in range(10_000):
    result = [f(t) for t in l]
end = time.perf_counter()
listCompTime = end - start
print(result)

start = time.perf_counter()
for _ in range(10_000):
    result = list(map(f, l))
end = time.perf_counter()
mapTime = end - start
print(result)

print(toEnggNotation(forLoopTime), toEnggNotation(listCompTime),
      toEnggNotation(mapTime))
```

2
2
2
3
2
4
2
5
2
6
2
7
2
8
2
9
3
0

```
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
12.5702m      0.012570190010592341 13.7350m      0.013735007989453152 16.9870m      0.0169869739911519
[1, 1, 2, 4, 8, 9, 27, 81, 64, 256]
12.5702m      0.012570190010592341 13.7350m      0.013735007989453152 16.9870m      0.0169869739911519
```

In [349]:

```
def power(x, y):
    return x ** y

def dist(pt1, pt2):
    return math.sqrt(sum(coord1 - coord2 for coord1, coord2 in zip(pt1,
pt2)))

def partial(f, operand2):

    def calcPartialFunction(operand1):
        return f(operand1, operand2)

    return calcPartialFunction

squares = partial(power, 2)
print(squares(3))
print(squares(4))
dist_from_origin = partial(dist, (0, 0))
print(dist_from_origin((1, 1)))
gcd_60 = partial(math.gcd, 60)
print(gcd_60(13))
log_2 = partial(math.log, 2)
print(log_2(8))
log_10 = partial(math.log, 10)
```

```
4 print(log_10(100))
1 log_16 = partial(math.log, 16)
5 print(log_16(65536))
```

```
9
16
1.4142135623730951
1
3.0
2.0
4.0
```

In [350]:

```
1 def norm(x, y):
2     return math.sqrt(x ** 2 + y ** 2)
3
4
5 def find_index_min(seq):
6     minima = min(seq)
7     return seq.index(minima)
8
9
10 def logged(f):
11
12     def g(*args):
13         print("Logging...")
```

```

14         start = time.perf_counter()
15         print(f(*args))
16         end = time.perf_counter()
17         print(toEnggNotation(end - start))
18         print("... Finished")
19
20     return g
21
22
23 loggedNorm = logged(norm)
24 loggedNorm(3, 4)
25 loggedFindIndexMin = logged(find_index_min)
26 loggedFindIndexMin([8, 7, 2, 9, 4])

```

```

Logging...
5.0
18.7050µ      1.8705002730712295e-05
... Finished
Logging...
2
11.0510µ      1.1050986358895898e-05
... Finished

```

```

In [351]: 1 def encloseGetPower(f):
2
3           def getPower(*args):
4               return f(*args)
5
6           print(hex(id(getPower)))
7           return getPower
8
9
10 getPower = encloseGetPower(pow)
11 print(getPower(2, 5))
12 print(hex(id(getPower)))
13 print(hex(id(encloseGetPower)))
14 print(getPower.      )
15 print(encloseGetPower.      )

```

```

0x74b66f154930
32
0x74b66f154930
0x74b66f155010
(<cell at 0x74b67208d690: builtin_function_or_method object at 0x74b6ccd3c630>,)
None

```

```

In [352]: 1 l = [(1, 2), (-4, -5.5), (10, -10), (-2, 2)]
2          print(sorted(l, key = lambda t: abs(t[0] - t[1]), reverse = True))

```

```

[(10, -10), (-2, 2), (-4, -5.5), (1, 2)]

```

```

In [353]: 1 l = ["s", "0", "1"]
2          card = "".join(reversed(l))
3          print(card)
4          card = ""

```

```

5 for k in reversed(l):
6     card += k
7 print(card)

```

10s

10s

In [354]:

```

1
2
3
4
5 suits = 'shdc' # Spades, Hearts, Diamonds, Clubs
6 ranks = list('23456789') + ['10', 'J', 'Q', 'K', 'A']
7 cards = [[rank + suit for rank in ranks] for suit in suits]
8
9
10 def sorter(suits, reverse = False):
11     return sorted(suits, key = lambda card:
12 ranks.index(card.strip("s").strip("h").strip("d").strip("c")), reverse =
13 reverse)
14
15
16 sortedCards = []
17 for suit in cards:
18     sortedCards.append(sorter(suit, reverse = True))
19 print(sortedCards)
20
21
22 sortedCards = []
23 for suit in cards:
24     sortedCards.append(sorter(suit))
25 print(sortedCards)
26
27
28
29
30
31
32
33
34
35
36
37
38

```

```

[['As', 'Ks', 'Qs', 'Js', '10s', '9s', '8s', '7s', '6s', '5s', '4s', '3s', '2s'], ['Ah', 'Kh', 'Qh', 'Jh', '10h', '9h', '8h', '7h', '6h', '5h', '4h', '3h', '2h'], ['Ad', 'Kd', 'Qd', 'Jd', '10d', '9d', '8d', '7d', '6d', '5d', '4d', '3d', '2d'], ['Ac', 'Kc', 'Qc', 'Jc', '10c', '9c', '8c', '7c', '6c', '5c', '4c', '3c', '2c']]
[['2s', '3s', '4s', '5s', '6s', '7s', '8s', '9s', '10s', 'Js', 'Qs', 'Ks', 'As'], ['2h', '3h', '4h', '5h', '6h', '7h', '8h', '9h', '10h', 'Jh', 'Qh', 'Kh', 'Ah'], ['2d', '3d', '4d', '5d', '6d', '7d', '8d', '9d', '10d', 'Jd', 'Qd', 'Kd', 'Ad'], ['2c', '3c', '4c', '5c', '6c', '7c', '8c', '9c', '10c', 'Jc', 'Qc', 'Kc', 'Ac']]

```

In [355]:

```
1
2
3 data = {'S1': (100, 200, 80, 180), 'S2': (10, 20, 8, 18), 'S3': (50, 150, 50,
4 150)}
5
6 def findMinima(data, reverse = False):
7     minimax = min if reverse else max
8     return minimax(data, key = lambda d: data[d][1] - data[d][2])
9
10 print(findMinima(data, reverse = True))
11 print(findMinima(data))
12
```

S2
S1

In [356]:

```
1 quotes = [('AACC', 6.05, 6.07, 6.03, 6.05, 65800), ('AAME', 1.7, 1.82, 1.7,
2 1.82, 4300), ('AAON', 24.98, 25.07, 24.9, 24.94, 28200), ('AAPL', 317.99,
3 319.57, 316.75, 317.13, 12901800), ('AATI', 3.82, 3.82, 3.74, 3.79, 194600),
4 ('AAWW', 60.89, 61.44, 60.5, 61.19, 272800), ('AAXJ', 65.4, 65.71, 65.28,
5 65.56, 390300), ('ABAT', 4.01, 4.01, 3.95, 3.99, 656300), ('ABAX', 25.26,
6 25.49, 25.04, 25.42, 73700), ('ABBC', 11.75, 11.88, 11.48, 11.53, 29700),
7 ('ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCD', 3.25, 3.25, 3.11, 3.22,
8 122800), ('ABCO', 48.75, 50.41, 46.9, 50.37, 66300), ('ABCW', 0.52, 0.61, 0.52,
9 0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABIO', 3.96, 4,
10 3.88, 4, 38500), ('ABMD', 11.94, 12, 11.69, 11.87, 122600), ('ABTL', 0.82,
11 0.84, 0.82, 0.83, 28700), ('ABVA', 3.09, 3.25, 3.09, 3.25, 6200), ('ACAD',
12 0.76, 0.76, 0.7, 0.74, 341500), ('ACAS', 7.52, 7.72, 7.52, 7.66, 5199800),
13 ('ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1,
14 456100), ('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5,
15 1.6, 12300), ('ACFN', 3.82, 4, 3.82, 3.98, 53700), ('ACGL', 89.76, 90.14,
16 89.39, 89.92, 240900), ('ACGY', 22.41, 22.56, 22.25, 22.46, 86800), ('ACHN',
17 3.12, 3.2, 3.07, 3.16, 113700), ('ACIW', 26.96, 27.03, 26.63, 26.8, 157000),
18 ('ACLI', 33.65, 33.77, 33.45, 33.63, 28700), ('ACLS', 2.47, 2.63, 2.46, 2.53,
19 1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71, 158600), ('ACOM', 25.2, 26.6, 24.9,
20 26.56, 265300), ('ACOR', 26.67, 27.07, 26.38, 27.04, 1415000), ('ACPW', 1.84,
21 1.89, 1.77, 1.85, 565500), ('ACTG', 27.2, 27.43, 26.86, 27.18, 228800),
22 ('ACTI', 3.25, 3.26, 3.25, 3.26, 148500), ('ACTS', 2.08, 2.09, 2.07, 2.07,
23 130500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ACWI', 46.53, 46.7, 46.32,
24 46.51, 286200), ('ACWX', 44.49, 44.66, 44.36, 44.6, 55500), ('ACXM', 18, 18.07,
25 17.81, 18.01, 289800), ('ADAM', 7.34, 7.49, 7.33, 7.44, 81700), ('ADAT', 0.6,
26 0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300),
27 ('ADCT', 12.68, 12.69, 12.66, 12.68, 1660500), ('ADEP', 6.14, 6.14, 4.95, 5.61,
28 71000), ('ADES', 6.2, 6.22, 6, 6.19, 4800), ('ADGF', 4.31, 4.55, 4.31, 4.54,
```

```
10200)]
```

```
def myFilter(percentage = 10):  
    return list(filter(lambda t: abs((t[4] - t[2]) / t[2]) > (percentage /  
100), quotes))  
  
print("10:", myFilter(10))  
print("5:", myFilter(5))  
print("1:", myFilter(1))
```

```
10: [('ABCW', 0.52, 0.61, 0.52, 0.53, 83000)]  
5: [('ABCW', 0.52, 0.61, 0.52, 0.53, 83000), ('ACFC', 1.69, 1.7, 1.5, 1.6, 12300), ('ADEP', 6.14,  
6.14, 4.95, 5.61, 71000)]  
1: [('ABBC', 11.75, 11.88, 11.48, 11.53, 29700), ('ABCB', 9.3, 9.3, 9.06, 9.14, 42600), ('ABCW',  
0.52, 0.61, 0.52, 0.53, 83000), ('ABFS', 25.98, 26.27, 25.41, 25.5, 384900), ('ABMD', 11.94, 12,  
11.69, 11.87, 122600), ('ABTL', 0.82, 0.84, 0.82, 0.83, 28700), ('ACAD', 0.76, 0.76, 0.7, 0.74,  
341500), ('ACAT', 14.44, 14.44, 14.04, 14.2, 51700), ('ACCL', 8.11, 8.21, 7.94, 8.1, 456100),  
('ACET', 8.01, 8.04, 7.13, 7.73, 575600), ('ACFC', 1.69, 1.7, 1.5, 1.6, 12300), ('ACHN', 3.12, 3.2,  
3.07, 3.16, 113700), ('ACLS', 2.47, 2.63, 2.46, 2.53, 1818800), ('ACMR', 2.69, 2.84, 2.37, 2.71,  
158600), ('ACPW', 1.84, 1.89, 1.77, 1.85, 565500), ('ACUR', 2.6, 2.64, 2.51, 2.6, 16000), ('ADAT',  
0.6, 0.68, 0.59, 0.66, 86400), ('ADBE', 29.43, 29.71, 29.07, 29.14, 7585300), ('ADEP', 6.14, 6.14,  
4.95, 5.61, 71000)]
```

```
In [357]: 1 l = [5, 6, -4, 8]  
          2 print(min(l, key = abs))
```

```
-4
```

```
In [358]: 1 importlib.reload(logging)  
          2 logging.basicConfig(format = "%(asctime)s %(levelname)s %(message)s", datefmt =  
          3 "%FT%Tb")  
          logging.error("This is an error")
```

```
2026-09-14T03:33:48b ERROR This is an error
```

```
In [359]: 1 logger = logging.getLogger(  
          2 logger.setLevel(logging.DEBUG)  
          3 logger.error("This is an error message")  
          4 logger.debug("This is a debug message")
```

```
2026-09-14T03:33:48b ERROR This is an error message  
2026-09-14T03:33:48b DEBUG This is a debug message  
2026-09-14T03:33:48b DEBUG This is a debug message
```

```
In [360]: 1 def f():  
          2     pass  
          3  
          4  
          5 print(f.  
          6 print(type(f.
```

```
f  
<class 'str'>
```

In [361]:

```
1 def partial(f):
2
3     def callF(*pargs, **kwargs):
4         logger.debug("Calling your function... " + f.          )
5         start = time.perf_counter()
6         result = f(*pargs, **kwargs)
7         end = time.perf_counter()
8         logger.debug("... Function returned.")
9         logger.info(f"Elapsed Time: {toEnggNotation(end - start)}")
10        # print(f"{toEnggNotation(end - start)}")
11        return result
12
13    return callF
14
15
16 def add(a, b):
17     return a + b
18
19
20 def log3(a):
21     return math.log(a, 3)
22
23
24 @partial
25 def log4(a):
26     return math.log(a, 4)
27
28
29 myAdder = partial(add)
30 logger.info("myAdder: " + str(myAdder(2, 3)))
31 log3 = partial(log3)
32 logger.info("log3: " + str(log3(9)))
33 log4 = partial(log4)
```

```

2026-09-14T03:33:49b DEBUG Calling your function... add
2026-09-14T03:33:49b DEBUG ... Function returned.
2026-09-14T03:33:49b INFO Elapsed Time: 1.6130μ 1.613021595403552e-06
2026-09-14T03:33:49b INFO myAdder: 5
2026-09-14T03:33:49b DEBUG Calling your function... log3
2026-09-14T03:33:49b DEBUG ... Function returned.
2026-09-14T03:33:49b INFO Elapsed Time: 4.3880μ 4.3879845179617405e-06
2026-09-14T03:33:49b INFO log3: 2.0
2026-09-14T03:33:49b DEBUG Calling your function... log4
2026-09-14T03:33:49b DEBUG ... Function returned.
2026-09-14T03:33:49b INFO Elapsed Time: 4.1980μ 4.197994712740183e-06
2026-09-14T03:33:49b INFO log4: 2.0
2026-09-14T03:33:49b DEBUG Calling your function... log3
2026-09-14T03:33:49b DEBUG ... Function returned.
2026-09-14T03:33:49b INFO Elapsed Time: 4.3880μ 4.3879845179617405e-06
2026-09-14T03:33:49b INFO log3: 2.0
2026-09-14T03:33:49b DEBUG Calling your function... log4
2026-09-14T03:33:49b DEBUG ... Function returned.
2026-09-14T03:33:49b INFO Elapsed Time: 4.1980μ 4.197994712740183e-06
2026-09-14T03:33:49b INFO log4: 2.0

```

In [362]:

```

1  def cache(f):
2      print("Initializing cache...")
3      cacheDict = {}
4
5      def callF(*pargs):
6          if pargs in cacheDict:
7              print("Cache Hit")
8              return cacheDict[pargs]
9          else:
10             print("Cache Miss")
11             result = f(*pargs)
12             cacheDict[pargs] = result
13             return result
14
15     return callF
16
17
18 @cache
19 def add(a, b):
20     return a + b
21
22
23 @cache
24 def mul(a, b):
25     return a * b

```

Initializing cache...

Initializing cache...

In [363]:

```

1  print(add.      )
2  print(mul.      )

```

```
(<cell at 0x74b66f141fc0: dict object at 0x74b66ef93340>, <cell at 0x74b66f143940: function object at 0x74b66ef8bab0>)  
(<cell at 0x74b66f141c90: dict object at 0x74b66ef933c0>, <cell at 0x74b66f142260: function object at 0x74b66ef8bcc0>)
```

```
In [364]: 1 print(add(2, 3))  
          2 print(mul(2, 3))  
          3 print(add(2, 3))  
          4 print(mul(2, 3))
```

```
Cache Miss  
5  
Cache Miss  
6  
Cache Hit  
5  
Cache Hit  
6
```

```
In [365]: 1 print(type(functools.lru_cache))
```

```
<class 'function'>
```

```
In [366]: 1 @functools.lru_cache(maxsize = 2)  
          2 def graham(a, b, c):  
          3     print("Cache Miss")  
          4     return a ** b ** c  
          5  
          6  
          7 print(graham(2, 2, 4))
```

```
Cache Miss  
65536
```

```
In [367]: 1 print(graham(2, 2, 4))  
          2 print(graham(2, 2, 4))  
          3 print(graham(2, 2, 5))  
          4 print(graham(2, 2, 6))  
          5 print(graham(2, 2, 4))
```

```
65536  
65536  
Cache Miss  
4294967296  
Cache Miss  
18446744073709551616  
Cache Miss  
65536
```

```
In [368]: 1 @functools.lru_cache(maxsize = 100)  
          2 def fib(n):  
          3     print(f"Calling fib{n}...")  
          4     if n <= 1:  
          5         return n  
          6     return fib(n - 1) + fib(n - 2)  
          7
```

8

9

```
print(fib(2000))
```

Calling fib2000...
Calling fib1999...
Calling fib1998...
Calling fib1997...
Calling fib1996...
Calling fib1995...
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204312082516817125
```

In [369]:

```
1 def myTimer(f):
2
3     def callF(*args):
4         start = time.perf_counter()
5         result = f(*args)
```

```

6         end = time.perf_counter()
7         print(toEnggNotation(end - start))
8         return result
9
10        return callF
11
12
13    # timedPower = myTimer(math.pow)
14    # print(timedPower(5, 188))
15    #
16    #
17    @myTimer
18    def myPower(a, b):
19        return a ** b
20
21
22    print(myPower(5, 188))

```

```

2.4240µ 2.4239998310804367e-06
2548947057811923643246208636428338888945767288308190057718217365417860642420893057508943799552292297
50557351508177816867828369140625

```

```
In [370]: 1 PRECISION = 4
```

```
In [371]: 1 a = decimal.Decimal(math.e)
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.718281828459045090795598298427648842334747314453125
<class 'decimal.Decimal'>
2.718281828459045

```

```
In [372]: 1 a = decimal.Decimal("2.404040313893824388847472828")
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.404040313893824388847472828
<class 'decimal.Decimal'>
2.4040403138938244

```

```
In [373]: 1 a = decimal.Decimal(2.2384892948592929275758693)
2 print(a)
3 print(type(a))
4 print(float(a))
```

```

2.23848929485929293292656439007259905338287353515625
<class 'decimal.Decimal'>
2.238489294859293

```

```
In [374]: 1 l = [decimal.Decimal(random.random()) for _ in range(10)]
2 a = sum(l) / len(l)
```

```
3 print(a)
4 print(type(a))
```

```
0.6115826496531003297896234017
<class 'decimal.Decimal'>
```

In [375]:

```
1 def normalize(f):
2
3     def callF(*args):
4         result = f(*args)
5         # print(type(result))
6         result = float(result)
7         return float(f"{result:.{PRECISION}f}")
8
9     return callF
```

In [376]:

```
1 @normalize
2 def perc_diff(x, y):
3     try:
4         return (y - x) / x * 100
5     except ZeroDivisionError:
6         return 0
7
8
9 @normalize
10 def sum_squares(*args):
11     return sum(x ** 2 for x in args)
12
13
14 @normalize
15 def avg(*args):
16     if len(args) == 0:
17         return 0
18
19     numbers = [decimal.Decimal(x) for x in args]
20     sum_ = sum(numbers)
21     return sum_ / len(numbers)
```

In [377]:

```
1 perc_diff(5088, 2872)
```

Out[377]: -43.5535

In [378]:

```
1 avg(2.4, 4.1, 5.932929, 8.2838485)
```

Out[378]: 5.1792

In [379]:

```
1 def add(x, y):
2     time.sleep(2)
3     return x + y
4
```

```
5 print(timeit.timeit("add(2340320832580385043843805805,  
6 320342408320482804320584305)", globals = globals(), number = 10))
```

20.00085302401567

In [380]:

```
1 @functools.lru_cache  
2 def add(x, y):  
3     time.sleep(2)  
4     return x + y  
5  
6  
7 print(timeit.timeit("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

2.000089545006631

In [381]:

```
1 def add(x, y):  
2     time.sleep(2)  
3     return x + y  
4  
5  
6 print(timeit.repeat("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

[20.000788189005107, 20.001294915011385, 20.000837992003653, 20.000805342016974, 20.000813703984022]

In [382]:

```
1 @functools.lru_cache  
2 def add(x, y):  
3     time.sleep(2)  
4     return x + y  
5  
6  
7 print(timeit.repeat("add(2340320832580385043843805805,  
320342408320482804320584305)", globals = globals(), number = 10))
```

[2.000094559014542, 1.613021595403552e-06, 8.019851520657539e-07, 7.110065780580044e-07, 6.920017767697573e-07]

In [383]:

```
1 def graham(x, y, z):  
2     return x ** y ** z  
3  
4  
5 print(toEnggNotation(timeit.timeit("graham(4, 5, 6)", globals = globals(),  
number = 10)))
```

519.2740µ 0.0005192740063648671

In [384]:

```
1 @functools.lru_cache  
2 def graham(x, y, z):  
3     return x ** y ** z  
4  
5  
6 print(toEnggNotation(timeit.timeit("graham(2, 3, 4)", globals = globals(),  
number = 10)))
```

3.4870μ 3.4869881346821785e-06

```
In [385]: 1 print(toEnggNotation(timeit.timeit("2 ** 3 ** 4", number = 100)))
```

19.2760μ 1.9275990780442953e-05

```
In [386]: 1 def normalize(precision):
2
3         def callF(f):
4
5             def callFInner(*args):
6                 result = f(*args)
7                 result = f"{result:.{precision}f}"
8                 return result
9
10            return callFInner
11
12        return callF
```

```
In [387]: 1 @normalize(2)
2 def perc_diff(x, y):
3     try:
4         return (y - x) / x * 100
5     except ZeroDivisionError:
6         return 0
7
8
9 @normalize(4)
10 def sum_squares(*args):
11     return sum(x ** 2 for x in args)
12
13
14 @normalize(8)
15 def avg(*args):
16     if len(args) == 0:
17         return 0
18
19     numbers = [decimal.Decimal(x) for x in args]
20     sum_ = sum(numbers)
21     return sum_ / len(numbers)
```

```
In [388]: 1 print(perc_diff(3, 7))
2 print(sum_squares(3, 4, 5))
3 print(avg(1.25, 1.75, 1.5, 1.0, 2.0, 2.5, 2.25, 2.75))
```

133.33
50.0000
1.87500000

```
In [389]: 1 def normalize(f):
2
3         def callF(*args):
```

```
4         result = f(*args)
5         return result
6
7     return callF
```

In [394]:

```
1  try:
2
3      @normalize(2)
4      def perc_diff(x, y):
5          try:
6              return (y - x) / x * 100
7          except ZeroDivisionError:
8              return 0
9
10     @normalize(4)
11     def sum_squares(*args):
12         return sum(x ** 2 for x in args)
13
14     @normalize(8)
15     def avg(*args):
16         if len(args) == 0:
17             return 0
18
19         numbers = [decimal.Decimal(x) for x in args]
20         sum_ = sum(numbers)
21         return sum_ / len(numbers)
22
23 except TypeError as ex:
24     print(type(ex).      + ": " + ex.args[0] + ".")
```

TypeError: 'int' object is not callable.

In [509]:

```
@functools.lru_cache
1 def getFibonacci(i):
2     if i == 0:
3         return 1
4     if i == 1:
5         return 1
6     if i < 0:
7         return 0
8     else:
9         return getFibonacci(i - 1) + getFibonacci(i - 2)
10
11 print(getFibonacci(60))
12
13 with open("../var/out000.txt", "w") as fl:
14     fl.write("i,fib,fact,gcd\n")
15     for i in range(101):
16         fl.write(f"{i},{math.factorial(i)},{getFibonacci(i)},
17 {math.gcd(getFibonacci(i),math.factorial(i))}\n")
18
19 0
```

```
1 with open("../var/out000.txt") as fl:
1   for line in fl:
1       print(line.strip())
2
1
3
1
4
1
5
1
6
1
7
1
8
1
9
2
0
2
1
2
2
```

2504730781961
i,fib,fact,gcd
0,1,1,1
1,1,1,1
2,2,2,2
3,6,3,3
4,24,5,1
5,120,8,8
6,720,13,1
7,5040,21,21
8,40320,34,2
9,362880,55,5
10,3628800,89,1
11,39916800,144,144
12,479001600,233,1
13,6227020800,377,13
14,87178291200,610,10
15,1307674368000,987,21
16,20922789888000,1597,1
17,355687428096000,2584,136
18,6402373705728000,4181,1
19,121645100408832000,6765,165
20,2432902008176640000,10946,26
21,51090942171709440000,17711,1
22,112400072777607680000,28657,1
23,25852016738884976640000,46368,46368
24,620448401733239439360000,75025,25
25,1551121004330985984000000,121393,1
26,403291461126605635584000000,196418,34
27,10888869450418352160768000000,317811,39
28,304888344611713860501504000000,514229,1
29,8841761993739701954543616000000,832040,440
30,265252859812191058636308480000000,1346269,1
31,8222838654177922817725562880000000,2178309,21
32,263130836933693530167218012160000000,3524578,2
33,8683317618811886495518194401280000000,5702887,1
34,295232799039604140847618609643520000000,9227465,65
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```

066272908038299975680000000000000000000, 2880067194370816120, 268748920
90, 1485715964481761497309522733620825737885569961284688766942216863704985393094065876545992131370884
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18286253697920827223758251185210916864000000000000000000000, 573147844013817084101, 1

```

In []:

```

fibonacciNumbers = {}
1 factorialNumbers = {}
gcd = {}
2
with open("../var/out000.txt") as fl:
3     next(fl)
    for ln in fl:
4         l = ln.strip().split(",")
            factorialNumbers[int(l[0])] = int(l[1])
5            fibonacciNumbers[int(l[0])] = int(l[2])
            gcd[(int(l[1]), int(l[2]))] = int(l[3])
6
    # print(factorialNumbers)
7    # print(fibonacciNumbers)
    # print(gcd)
8
9    # @functools.lru_cache
1   def getFibonacci(i):
    # print(f"getFibonacci({i}):")
    if i < 0:
1         return 0
    if i in fibonacciNumbers:
1         print(f"getFibonacci({i}): Found in cache!")
            return fibonacciNumbers[i]
2         print("Not found in cache...")
    fibonacciNumbers[i] = getFibonacci(i - 1) + getFibonacci(i - 2)
3
1         # return fibonacciNumbers.get(i, getFibonacci(i - 1) + getFibonacci(i
5         - 2))
1

```

```

6
1 print(getFibonacci(110))
7 print(getFibonacci(11))
1
8 # print(fibonacciNumbers)
1
9
2 def f(i):
0     print("f:", i)
2
1     def g():
2         return math.factorial(i)
2
2     return g
3
2
4 def getFactorial(i):
2     if i in factorialNumbers:
5         return factorialNumbers[i]
2     print("Not found in cache...")
6     return f(i)()
2     # return factorialNumbers.get(i, math.factorial(i))
7     # return factorialNumbers.get(i, f(i)())
2
8
2 print(getFactorial(110))
9 print(getFactorial(11))
3
0 # print(math.factorial(110))
3
1 # print(gcd)
3
2
3 def f1(i, j):
3     print("f1:", i, j)
3
4     def g1():
3         return math.gcd(i, j)
5
3     return g1
6
3
7 # print("getting:", gcd.get((5040, 21)))
3
8
3 def getGCD(i, j):
9     if (i, j) in gcd:
4         return gcd[(i, j)]
0     print("Not found in cache... Calling math.gcd():")
4     return math.gcd(i, j)
1
4     # return gcd.get((i, j), f1(i, j)())

```

2
4
3
4
4
4
5
4
6
4
7
4
8
4
9
5
0
5
1
5
2
5
3
5
4
5
5
5
6
5
7
5
8
5
9
6
0
6
1
6
2
6
3
6
4
6
5
6
6
6
7
6

```
print(getGCD(5040, 21))  
print(getGCD(5040, 22))
```

8	6
9	7
0	7
1	7
2	7
3	7
4	7
5	7
6	7
7	7
8	7
9	8
0	8
1	8
2	8
3	8
4	8
5	8
6	8

70492524767089125814114

144

```
Not found in cache...
```

f: 110

1588245541522742940425370312709077287172441023447356320758174831844456716294818303095996013151767852

047924367263817999020852114862342226687675762391121920000000000000000000000000

39916800

21

```
Not found in cache... Calling math.gcd():
```

2